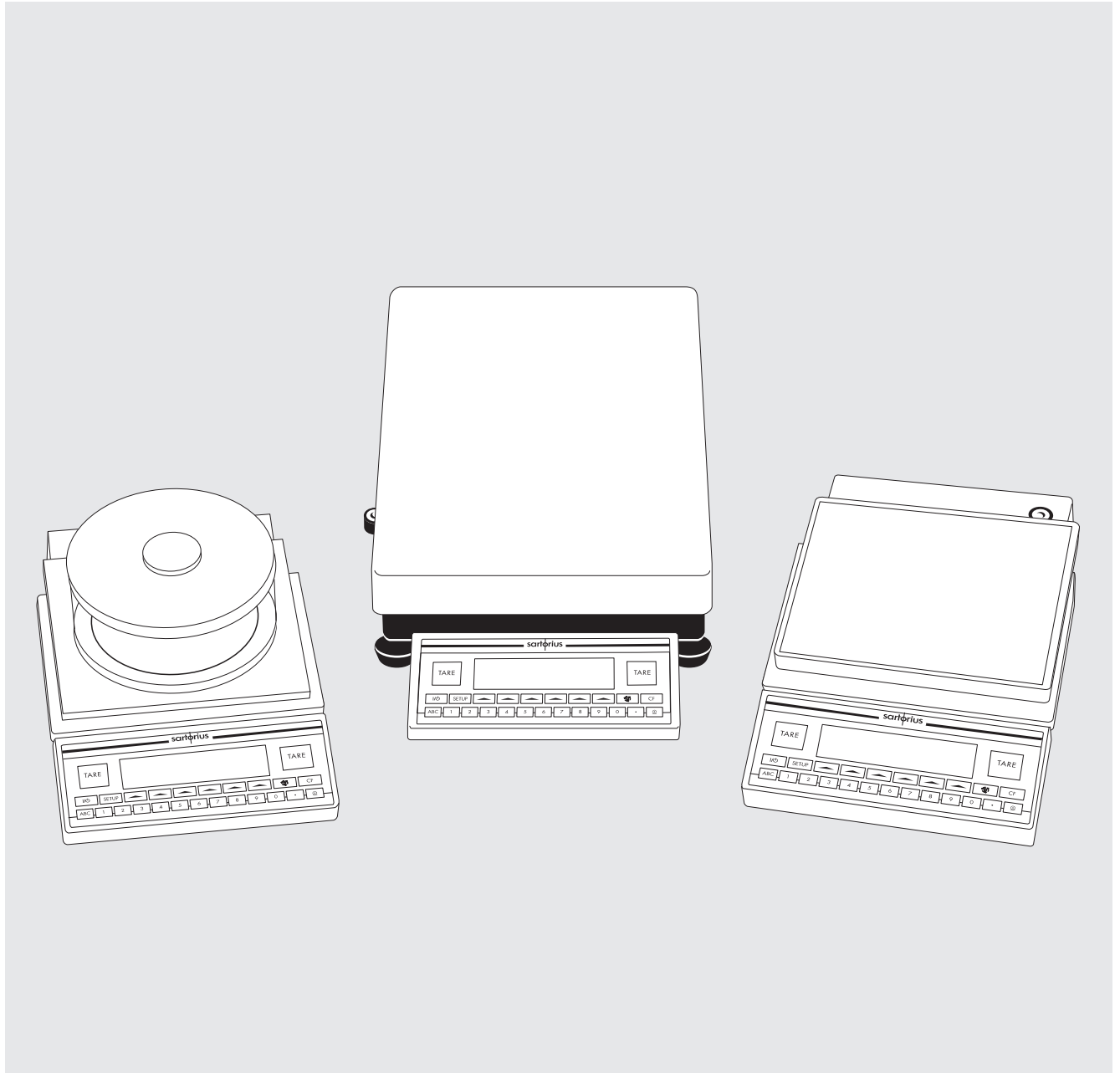


Sartorius Master^{pro} Serie

Electronic Precision Balances
Service Manual
for LP / LA Models



Sartorius AG, Weighing Technology

sartorius

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General Information

Note:

Repairs

Prerequisites for working on the Sartorius Master^{pro} Balances include considerable experience, completion of a Service Course at the main office in Goettingen, and the special tools and equipment described below. Please do not attempt or permit any unauthorized repair work!

If you remove the seal (warranty sticker), the balance is no longer covered under the warranty.

How to change the battery:

Important Note!

When exchanging the battery (if available), be sure that

- you use a new battery that is identical to the old one or equivalent to the battery recommended by the manufacturer.
- the poles are correctly positioned when soldering in the battery.

Warning: If you use the wrong battery, this may result in an explosion. Be sure to dispose of the defective battery properly according to your country's environmental protection laws!

Models

The LA/LP balance consists of a weighing cell (LA/LP) and a display and control unit.

LA/LP balances with brass systems:

LA/LP1 200S, LA/LP3 200D, LA/LP4 200S, LA/LP5 200P, LA/LP6 200S

LA/LP balances with small single-casting block systems:

LA/LP4 20, LA/LP2 20S, LA/LP6 20S, LA/LP6 20P, LA/LP8 20,
LA/LP2 200, LA/LP2 200S, LA/LP2 200P, LA/LP4 200, LA/LP6 200,
LA/LP1 2000S, LA/LP1 2000P

LA/LP balances with large single-casting block systems:

LA/LP1 6000S, LA/LP3 4000P, LA/LP3 4, LA/LP3 4001

Auxiliary Tools

All calibration/adjustment tasks can be performed without the service software, with the following exceptions:

- Span adjustment when the error is greater than 2%
- Calibration/adjustment after replacing a PCB

In addition to standard tools, you will need the following special tools:

Qty.	Designation	Order no.
1	SARTOCAS service software or	6740-33
1	PSION Server, complete	6739-98
1	RS-232 cable for connecting the balance to a PC	7357 312

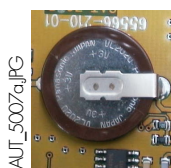
Accompanying Literature

Spare Parts List for the Master^{pro} (LA/LP) Balances
Operating Instructions for Electronic Precision (LA/LP) Balances
Sartorius SARTOCAS program description
Sartorius PSION Server program description
Sartorius Tool List - Status: 9/91

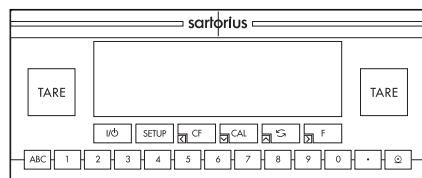
upright



flat



Operating the Balance



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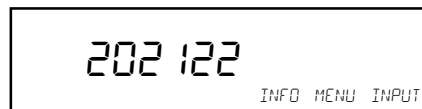
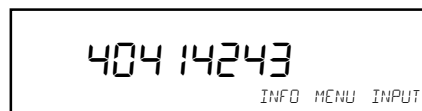
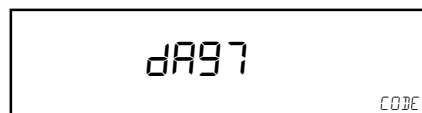
Key Functions

	On/off. Turns the display on/off. The balance remains in the standby mode.
	Clear function. Used to interrupt/cancel functions; e.g., delete data from memory, cancel calibration/adjustment functions, etc.
	Tare. Zeros the display.
	Print. The displayed data is output to the printer.
	Toggle. Toggles the readout between weight and calculated value when using the counting, weighing in percent or calculation programs; otherwise, toggles between two weight units.
	Function. Starts an application program
	SETUP. For accessing the SETUP program menu, storing menu codes, exiting the menu.
	Calibration/adjustment.
	Numeric input (e.g., ID number).
	Decimal point for numeric input.
	Toggle to alphabetic input.
	Arrow (cursor) keys.

Password Levels

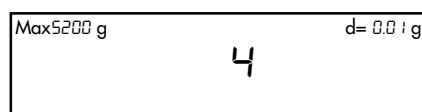
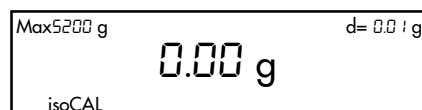
There are 3 different levels of password-protected access:

1. The user can assign a password to protect memory data
2. 40414243 is the "dealer password;" this can be given to the user if the user password is lost
3. 202122 is to be used only by service technicians; when this password is entered, the data entered using the INPUT mode can be changed and the key functions are expanded.



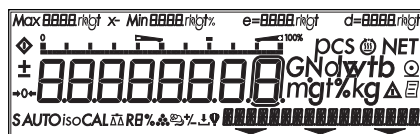
Setting the Language

- When 0.00 g is displayed (depending on balance model), you can select the language by entering the corresponding number (use the numeric keys).
 - 1 German
 - 2 English (factory setting)
 - 3 English with US date/time format
 - 4 French
 - 5 Italian
- Press to save the setting.

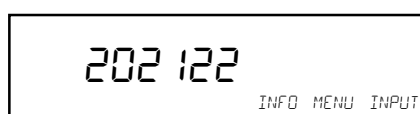
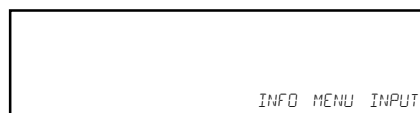


Example:

French: Press 4 and then press to store.



LP_anzei.EPS



Note:

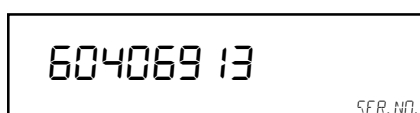
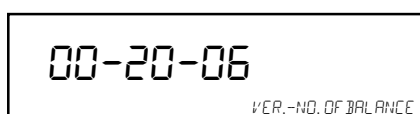
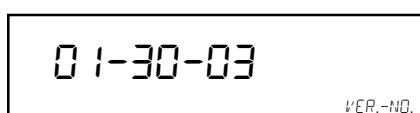
SETUP Program

- Press **[I/O]** to turn on the balance.
- Wait until the power-up routine is completed (delayed power-on, display segment test, function test).
- When 0.00 g is displayed (depending on the balance model), press **[SETUP]**.
- The 3 soft key functions are displayed.
- Press **[CF]** to access the "INFO" mode.
- Press **[S]** to access the "MENU."
- Press **[F]** to access the "INPUT" mode.
- If the password prompt (ENTER PASSW.) is displayed, a user password has been stored.
- Enter the service password (202 122) and press **[SETUP]** to confirm.
- An "S" is displayed in the lower left corner to indicate that the service mode is active.

The input options available in the MENU and in the INPUT mode are expanded when the balance is switched to the service mode (see "Overview of the User/Service SETUP Program," page 9).

INFO Mode

Select the INFO mode to display information about the balance.



Note:

- Press **[CAL]** while running the SETUP program.
- The version number of the display and control unit is displayed (in this example, 01 is the display and control unit, 30 the hardware and 03 the software).
- Press **[^]**.
- The version number of the processor for the weighing platform is displayed (in this example, 00 is the weighing platform, 20 the hardware and 06 the software).
- Press **[^]**.
- The model name is displayed (0000:) indicates that you can scroll the readout by pressing **[>]**.
- Press **[^]**.
- The balance serial number is displayed.
- Press **[Q]** to display information about the data output port.
- Press **[SETUP]** to exit the INFO mode.
- The balance returns to the weighing mode.

The data shown here are examples only; the data displayed on the balance you are working with will depend on the series, version and model of the balance.

INPUT Mode

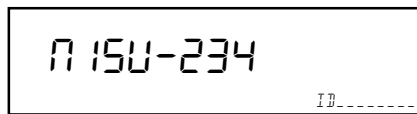
Press INPUT and then  to output the current settings to the data output port. Select the INPUT mode to enter data in the balance.

The input options available in the INPUT mode are expanded when the balance is switched to the service mode (e.g., for entering an external linearization weight).

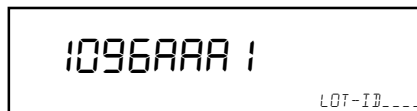
- Press  while running the SETUP program.

Note:

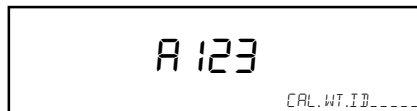
You can input up to 20 characters in an input field; use the left and right arrow keys ( / ) to scroll the display for input longer than 8 characters.


ID-----

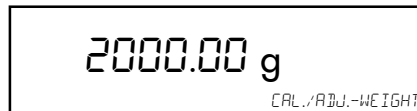
- Enter an ID number (e.g., inventory or laboratory number).


LOT-ID----

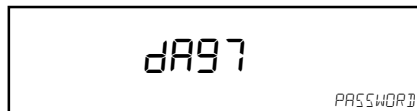
- Press  to select the individual positions.


CAL.WT.ID----

- Enter a lot number (e.g., test series number)
- Press 


CAL./ADJ.-WEIGHT

- Enter a calibration weight ID number (e.g., number of the weight set)
- Press 

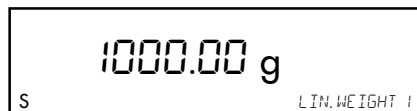

PASSWORD

- The calibration/adjustment weight is displayed. The weight shown can be changed, including the number of digits following the decimal point.
- Press 
- The *PASSWORD* is displayed (user password).
- Press 

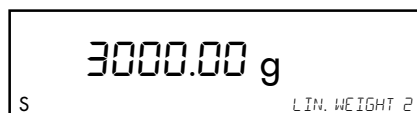
Note:

"S" is displayed if the service mode was activated before the INPUT mode was selected.

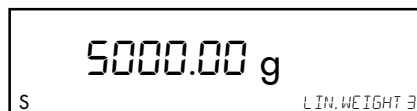
The following settings can only be changed in the service mode.


S LIN.WEIGHT 1

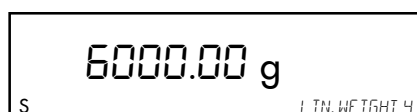
- Input for the 1st linearization weight. The weight shown can be changed, including the number of digits following the decimal point.


S LIN.WEIGHT 2

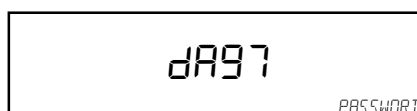
- Press 


S LIN.WEIGHT 3

- Input for the 2nd linearization weight. The weight shown can be changed, including the number of digits following the decimal point.


S LIN.WEIGHT 4

- Press 


PASSWORD

- Input for the 3rd linearization weight. The weight shown can be changed, including the number of digits following the decimal point.
- Press 

- Input for the 4th linearization weight. The weight shown can be changed, including the number of digits following the decimal point.
- Press 

- Input for the *PASSWORD* (user password)
This password protects the MENU and the INPUT mode.

The service password has higher priority than the user password.

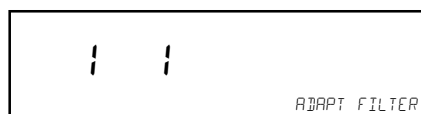


- Press **SETUP** to exit the INPUT mode.
- The balance returns to the weighing mode.

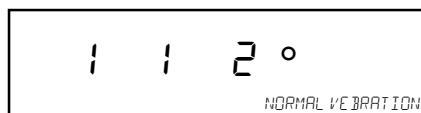


MENU - Balance Operating Menu

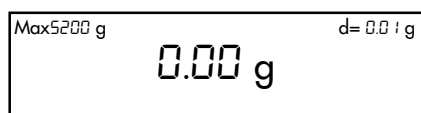
- Press **↺** while running the SETUP program.
- The 1st menu level is displayed.



- Use the arrow keys (**↓** **<** **^** **>**) to select the desired menu level and code digits.



- Press **>** to confirm a digit setting and move to the next digit.
- Press **SETUP** to store new code settings and exit the operating menu.

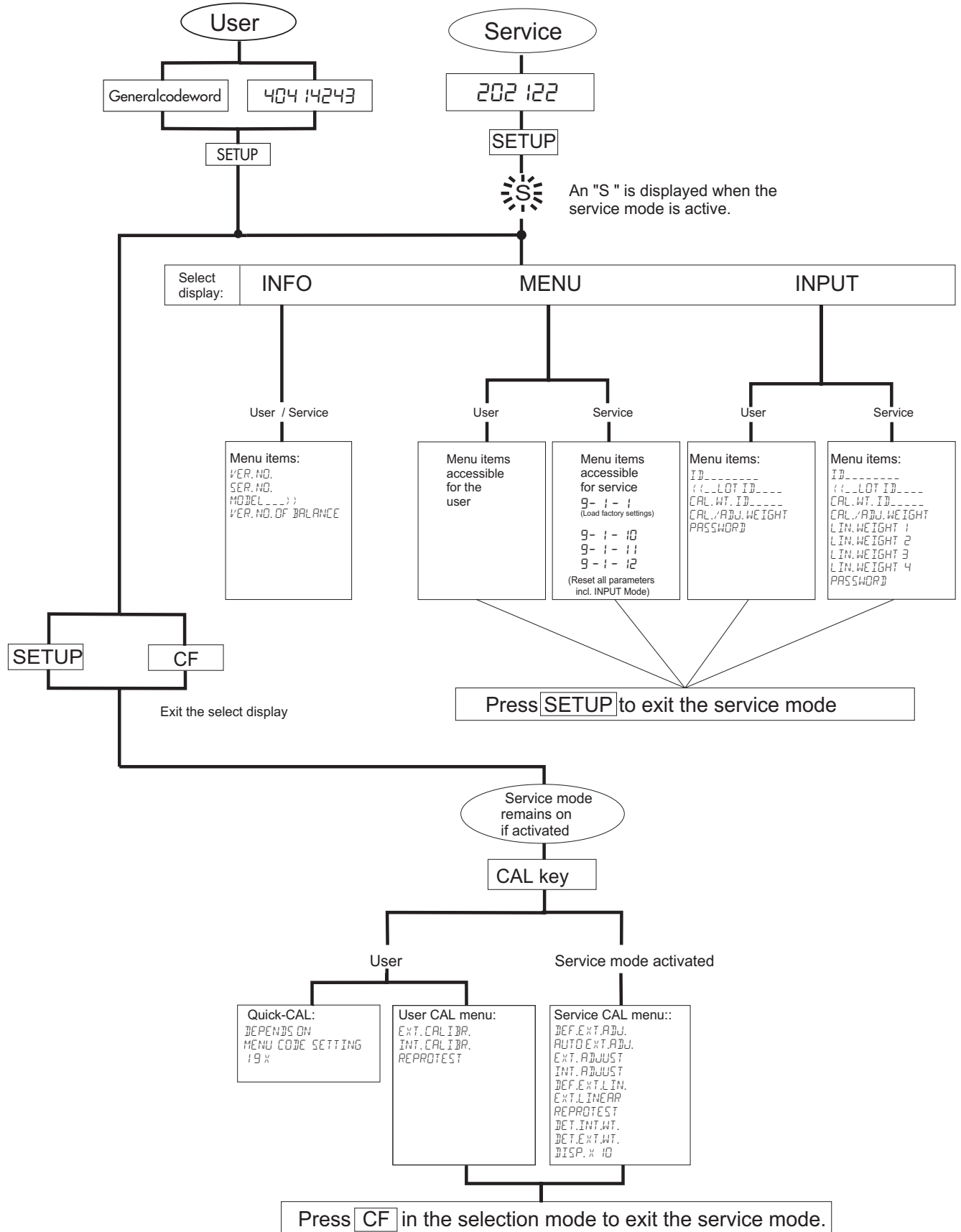


- The balance returns to the weighing mode.

Note:

See the section entitled "Balance Operating Menu Codes" starting on page 43 for an overview of the balance operating menu.

Overview of the User / Service SETUP Program

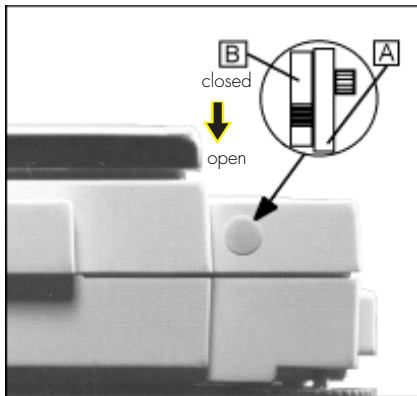


Service Switches

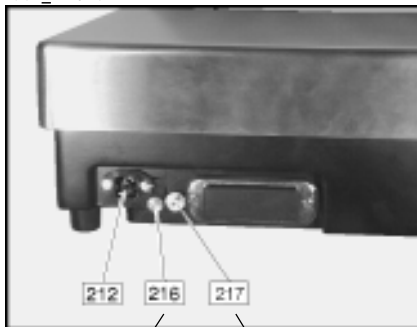
The Master^{pro} balances are equipped with a service mode; in this mode, both the SETUP functions and the calibration/adjustment functions are expanded.

Note:

Open the access switch (B) to calibrate/adjust the balance in the service mode (with the PSION Server or SARTOCAS Service Program for PCs).



WA_BPIO1.TIF



WA_BPIO2.TIF

Access Switch

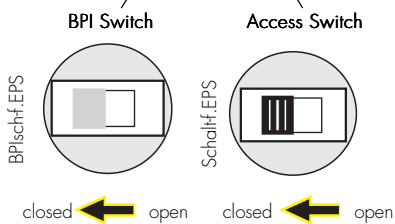
The access switch (B) and the BPI switch (A) are located under the cap on the rear panel of the balance, under a protective cap on the right side. On verified balances, this protective cap is sealed.

On high-capacity LP balances (≥ 16 kg), the access switch is located on the left, behind screw (217), and the BPI switch is behind the two-hole screw (216); both screws are next to the DC power jack (212).

Activating the BPI Mode / Deactivating the Write Protection

You must activate the BPI mode / deactivate the write protection:

- to calibrate/adjust the balance with the PSION Server, or
 - to replace the balance PCB.
-
- Turn on the balance.
 - When 0.00 g is displayed (depending on the balance model) remove the protective cap or the two-hole screw (216) and move the BPI switch (A).
 - The display goes blank; the line for metrological data begins flashing. Wait for this line to flash at least 3 times, and then return the BPI switch to the original position.
 - Wait for the display to return to the weighing mode.
 - Turn the balance off and then turn it back on again.
 - The balance is now ready to operate in the BPI mode (the write protection on the EEPROM has been deactivated).



Note:

When the balance is in the BPI mode, it is not possible to operate peripheral devices that use the RS-232 protocol.

"Err30" is displayed when you press .

Note:

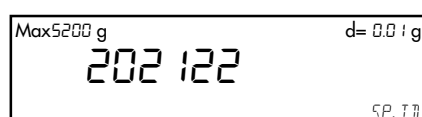
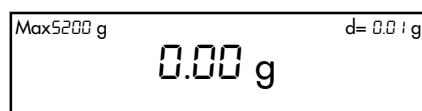
If the balance does not communicate with the service software in the BPI mode, you must remove the display and control unit (see page 17, "Removing the Display and Control Unit from the Weighing Cell").

Activating the Service Mode

The LP balances are equipped with a service mode that enables expanded calibration/adjustment functions. With these expanded functions, you do not need to use the PSION Server or SARTOCAS Service Program for PCs to calibrate/adjust the balance.

The service mode only remains active until:

1. you press **SETUP** to exit the menu, or
2. **CF** is pressed when the "Service CAL Selection" menu is displayed (see the overview on page 9).



The weighing mode is active (0.00 g is displayed - depending on the balance model)

- Enter the service password: 202 122.
- Press **SETUP**.
- The SETUP menu is displayed and an "S" is shown in the lower left corner.
- The service mode is now active.
- Press **CF** to exit the SETUP menu; press **CAL** to activate the calibration/adjustment menu.

Calibration/Adjustment in the Service Mode

Adjustment options in the service mode:

1. *INT. ADJUST ** – Span adjustment with internal weight
2. *DEF. EXT. LIN.* – External linearization with default weight
3. *EXT. LINEAR.* – External linearization (enter weight value in the INPUT mode)
4. *REPROTEST ** – Reproducibility test of the internal adjustment
5. *DET. INT. WT.* – Overwrite the internal calibration weight
6. *DET. EXT. WT.* – Determine and store the external weight (user-defined weight)
7. *DISP. % 10* – 10-fold increased display resolution; SuperRange resolution
8. *DEF. EXT. ADJ. ** – External span adjustment with default weight
9. *AUTO EXT. ADJ. ** – External span adjustment with automatic recognition of calibration weight
10. *EXT. ADJ. ** – External span adjustment with stored (user-defined) weight (entered in the INPUT mode)

Note:

In most cases it is not necessary to use service software programs (PSION or Sartocas) to adjust the LP balances, because all necessary adjustment functions are accessible when the balance is in the service mode.

You only need to use the adjustment software if the current adjustment error is greater than 2% or if multiple adjustment steps are necessary due to unfavorable ambient conditions.

* Can also be adjusted without activating the service mode

External Linearization

There are two ways to perform external linearization:

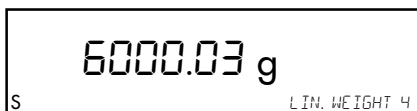
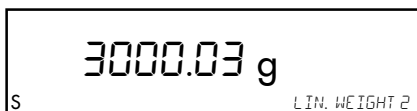
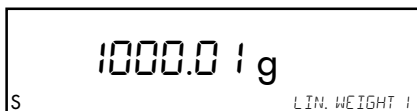
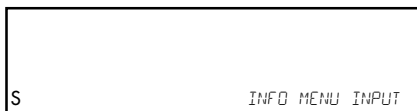
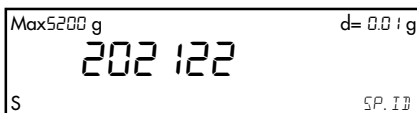
1. with the Sartorius service software (PSION Server or SARTOCAS Service Software for PCs) - if the adjustment error is greater than $\pm 2\%$, or
2. as described in this manual, with the balance in the service mode.

External Linearization with Manual Input of Weight Values

If the weights available differ greatly from the weights defined in the factory settings, you can precisely define the weights to be used:
(use only calibration weights!)

Manual Input of the Linearization Weights

- Press  to turn on the balance.
- The balance performs a self-test.
- 0.00 is displayed (depending on balance model)
- Activate the service mode:
 - Enter the service password: 202 122
 - Press .
- Press  to activate the INPUT mode.
- Press  (several times) to select the "LIN.WEIGHT 1" menu item and use the numeric keys to enter the calibration weight for the first linearization weight in accordance with the calibration certificate.
- Press  to store the weight value entered, and then press  to proceed to the "LIN.WEIGHT 2" menu item.
- Use the numeric keys to enter the calibration weight for the second linearization weight, press  to store, and then press  to proceed to the "LIN.WEIGHT 3" menu item.
- Use the numeric keys to enter the calibration weight for the third linearization weight, press  to store, and then press  to proceed to the "LIN.WEIGHT 4" menu item.
- Use the numeric keys to enter the calibration weight for the fourth linearization weight and press  to store.
- After defining all 4 linearization weights, press  to exit the INPUT mode.
- The balance returns to the weighing mode.



Performing External Linearization

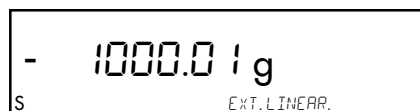
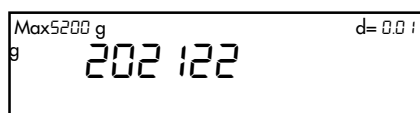
After defining the linearization weights, the linearization procedure can be started.

Note:

If linearization weights of sufficient accuracy are available (at least one digit greater than the display accuracy), it is not necessary to define the linearization weights (in the INPUT mode).

In this case, you can start linearization by selecting the "DEF.EXT.LIN." menu item.

The procedure is described below.



– Activate the service mode again (see page 11).

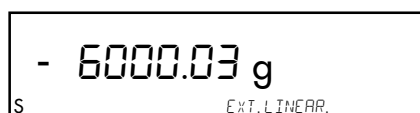
– Press [CAL].

– Press [F] (several times) to select the "EXT.LINEAR" menu item, tare the balance if necessary and press [CAL] to start the linearization routine.

– The first linearity weight is displayed.
Center the corresponding weight on the weighing pan.

Note:

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is performed automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last calibration is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).



– The deviation from the last adjustment is displayed.

– If the error exceeds the tolerance limits, either press [CAL] to make the corresponding adjustment, or press [CF] to cancel the adjustment procedure.

– The second linearity weight is displayed.

– Center the corresponding weight on the weighing pan and press [CAL] to confirm.

– Repeat this procedure until all linearity weights have been adjusted.

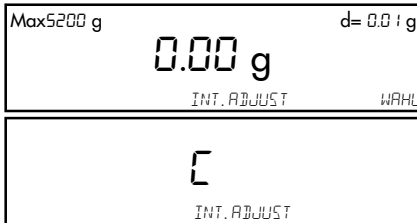
– After concluding the linearization routine, the service mode remains active until you press [CF].

Note:

Always perform internal or external span adjustment after external linearization.

Internal Span Adjustment

All LP balances are equipped with a built-in motorized calibration weight.



Note:

- Turn on the balance and allow it to warm up.
- Zero the balance, if necessary (press **[TARE]**).
- Press **[CAL]**; press **[F]** to select "INT. CALIBR." or "INT. ADJUST" and press **[CAL]** to start the internal span adjustment.
- The built-in calibration weight is applied automatically, the weight is stored, and the deviation from the last span adjustment is displayed.

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is made automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last calibration is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).



- If the error exceeds the tolerance limits, press **[CAL]** to make the corresponding adjustment, or press **[CF]** to cancel the adjustment procedure.
- The balance then returns to the SELECT mode.

External Span Adjustment

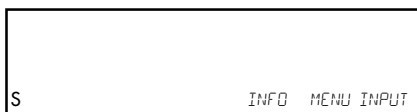
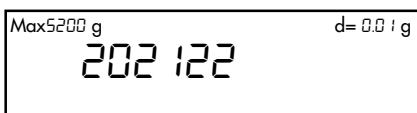
There are two ways to perform external span adjustment:

1. with the Sartorius service software (PSION Server or SARTOCAS Service Software for PCs) - if the adjustment error is greater than $\pm 2\%$, or
2. with the balance in the service mode, as described in this manual.

External Span Adjustment with Manual Input of Weight Values

You only need to input the calibration weight for span adjustment if:

1. the available weight is not sufficiently accurate (at least one digit greater than the display accuracy), or
2. you are working with a weight other than the recommended weight (e.g., with a heavier weight, close to the maximum capacity).



- Press **[I/O]** to turn on the balance.
- The balance performs a self-test.
- When 0.00 g is displayed (depending on balance model), enter the service password "202 122" and press **[SETUP]**.
- Press **[F]** to activate the INPUT mode.
- Press **[^]** (several times) to select the "CAL./ADJ. WEIGHT" menu item.
- Enter the conventional weight value, including all decimal places, using the numeric keys.
- Press **[SETUP]** to store the adjustment and exit the SETUP menu.

External Span Adjustment with Stored Weight



Note:

- Activate the service mode (see page 11).
- Press **[CAL]** and then press **[F]** to select the "USER EXT. ADJ." menu item.
- Tare the balance, if necessary, and then press **[CAL]** to start external span adjustment.
- Center the prompted calibration weight on the weighing pan.

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is made automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last calibration is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).

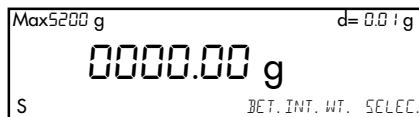
- If the error exceeds the tolerance limits, press **[CAL]** to make the corresponding adjustment, or press **[CF]** to cancel the adjustment procedure.
- The balance then returns to the SELECT mode.



Note:

Overwriting the Internal Calibration Weight

As a rule, this procedure is only necessary after the balance has been opened for service/repair work.

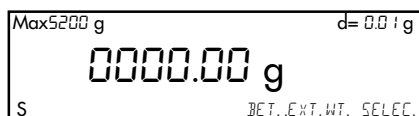


- Activate the service mode (see page 11).
- Perform external span adjustment (make sure you use a precise weight).
- Press **[F]** (several times) to select the "DET. INT. WT." menu item.
- Tare the balance, if necessary, and press **[CAL]** to overwrite the calibration weight.
- The built-in calibration weight is applied automatically, measured and written to the EEPROM in the balance processor.
- Press **[CF]** if you need to cancel this routine.
- The balance then returns to the SELECT mode.

Calibration with a User-defined Weight

You can define any weight as the calibration weight.

Activate the INPUT mode to enter the weight value and then store it under the "CAL. ADJ. WT." menu item.



- Activate the service mode (see page 11).
- Perform external or internal span adjustment.
- Press **[F]** to select the "CAL. ADJ. WT." menu item.
- Tare the balance if necessary, and then press **[CAL]** to start the calibration routine.
- The last calibration weight that was entered in the INPUT mode is displayed.
- Center the calibration weight you wish to define on the weighing pan.
- The weight value is displayed and stored.
- The balance returns to the SELECT mode.

reproTEST



- Activate the service mode (see page 11).
- Press **[F]** (several times) to select the "REPROTEST" menu item.

This tests the reproducibility of the automatic application of the built-in calibration weight.

The internal weight is applied, measured and removed 6 times.

At the end of the routine, the reproducibility value is displayed.

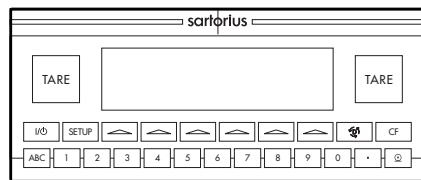
High Resolution - Display * 10



- Activate the service mode (see page 14).
- Press **[F]** (several times) to select the "DISP * 10" menu item.

The balance switches to SuperRange resolution and an additional digit is displayed (for test purposes and initial verification).

Operating the Balance



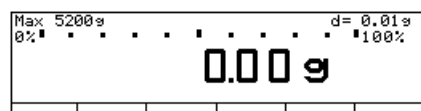
la_bedie.EPS

Key Functions

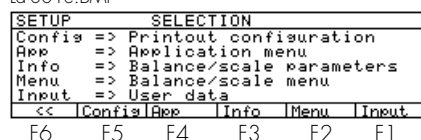
	On/off. Turns the display on/off. The balance remains in the standby mode.
	Clear function. Used to interrupt/cancel functions; e.g., delete data from memory, cancel calibration/adjustment functions, etc.
	Tare. Zeros the display.
	Print. The displayed data is output to the printer.
	Toggle. Toggles the readout between weight and calculated value when using the counting, weighing in percent or calculation programs; otherwise, toggles between two weight units.
	6 function keys that are defined individually depending on the program setting
	SETUP. For accessing the SETUP program menu, storing menu codes, exiting the menu.
	Numeric input (e.g., ID number).
	Decimal point for numeric input.
	Toggle to alphabetic input.

SETUP Program

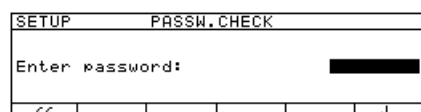
- Press to turn on the balance.
- Wait until the power-up routine is completed (delayed power-on, display segment test, function test).
- If the symbol to the left is displayed, press .
- The symbol to the left appears.
- The 6 function keys are automatically programmed with different functions. The function keys are numbered from right to left from 1 to 6, i.e. function key 1 is the key next to the key.
- Press F1 to access the "INPUT" mode.
- Press F2 to access the "MENU."
- Press F3 to access the "INFO" mode.
- Press F4 to access the "APPLIC (ATION)" menu
- Press F5 to access the "CONF IG (URATION)" menu
- Press F6 to return to the "WEIGHING" mode (the key has the same function).



la-001e.BMP



la-002e.BMP



la-003e.BMP

Note:

Note:

- If the symbol to the left is display after pressing a function key, a user password has been stored.
- This can be avoided by inputting a service password.
- In this case, input » 202 122 « and press F1 to confirm.

Except for the INFO mode, all other menus are stored by means of the user password.

The input options available in the MENU and in the INPUT mode are expanded when the balance is switched to the service mode (see "Overview of the User/Service SETUP Program," page 9).

Max 5200g	d= 0.01g
0%	100%
	3
	S ID

La-004e.BMP

Setting the Language

- When 0.00 g is displayed (depending on balance model), you can select the language by entering the corresponding number (use the numeric keys).
 - German
 - English (factory setting)
 - English with US date/time format
 - French
 - Italian
- Press **SETUP** to save the setting.

SETUP	SELECTION
Config =>	Printout configuration
App =>	Application menu
Info =>	Balance/scale parameters
Menu =>	Balance/scale menu
Input =>	User data
<<	Config App Info Menu Input

F5
La-002e.BMP

SETUP	CONFIG
Indiv. =>	Printout f. app./weighings
<<	Indiv.

La-005e.BMP

LIST	INDIV.PRT	SELECTION
		Blank line
		Form feed
		Date/time
		Time
<<	Delete	>

La-006e.BMP

SETUP	SELECTION
Config =>	Printout configuration
App =>	Application menu
Info =>	Balance/scale parameters
Menu =>	Balance/scale menu
Input =>	User data
<<	Config App Info Menu Input

F4
La-002e.BMP

SETUP	APPLICATION
Application 1 =>	Tossle wt.units
Application 2	Counting
Application 3	Percent weigh.
Extra func. (F4)	Animal weigh.
Extra func. (F5)	Calc., density
<<	Menu

La-007e.BMP

CONFIG Mode

Select the CONFIG(uration) mode to print out a record (on an interfaced printer).

Depending on the settings available in the **APPLI**(cation) menu, various configurations are available for printing out the record.

See the Installation and Operating Instructions for LP/LA Balances for a detailed description of the CONFIG mode.

APPLICATION Mode

The APPLI(cation) menu is used to input the settings and configuration for application programs.

Up to 3 application programs plus up to 2 additional functions can be activated at the same time.

See the Installation and Operating Instructions for LP/LA Balances for a detailed description of the setting options.

INFO Mode

Select the INFO mode to display information about the balance.

- Press F3 while running the SETUP program.
- The version number of the display and control unit, the version number of the processor for the weighing platform, the model name and the balance serial number are displayed.
- Press **F6** to exit the INFO mode or to exit the **SETUP** program, press **SETUP** (return to weighing mode).

SETUP	SELECTION
Config =>	Printout configuration
App =>	Application menu
Info =>	Balance/scale parameters
Menu =>	Balance/scale menu
Input =>	User data
<<	Config App Info Menu Input

F3
La-002e.BMP

SETUP	INFO
Version no.:	01-35-04
Bal. ver. no.:	00-20-06
Model:	LA5200P
Serial no.:	70404758
<<	

La-008e.BMP

Note:

The data shown here are examples only; the data displayed on the balance you are working with will depend on the series, version and model of the balance.

INPUT Mode

Select the INPUT mode to enter data in the balance.

The input options available in the INPUT mode are expanded when the balance is switched to the service mode (e.g., for entering an external linearization weight).

SETUP	SELECTION	S
Config =>	Printout configuration	
App =>	Application menu	
Info =>	Balance/scale parameters	
Menu =>	Balance/scale menu	
Input =>	User data	
<<	Config App Info Menu Input	

La-017e.BMP

F1

Note:

- Press **F1** while running the SETUP program.

You can input up to 20 characters in an input field; use the left and right arrow keys ($\leftarrow$ / $\rightarrow$) to scroll the display for input longer than 8 characters.

- The following user data can be displayed, entered or changed:

an ID number (e.g., inventory or laboratory number).

a lot number (e.g., measuring series)

a adjustment weight ID number (e.g., number of the adjustment weight)

The weight value of the calibration/adjustment weight, including the all necessary number of digits following the decimal point, can be changed using the numeric keys.

The weight value of the 4 linearization weights, including the all necessary number of digits following the decimal point, can be changed using the numeric keys!

Only accessible in the SERVICE mode!

Time

Date

SETUP	INPUT	S
Lin. wt. 2:	3000.00 g	
Lin. wt. 3:	5000.00 g	
Lin. wt. 4:	6000.00 g	
Time:	12.26.12	
Date:	08.12.97	
<<		

La-010e.BMP

SETUP	INPUT	S
Contrast(0-4):	2	
Password:		
ID1:		
ID2:		
ID3:		
<<		

La-011e.BMP

SETUP	INPUT	S
ID3:		
ID4:		
Adj. time1:		
Adj. time2:		
Adj. time3:		
<<		

La-012e.BMP

The contrast/angle of the display (input from 0 to 4, factory setting is 2)

Input an access password (user password)

This password stores all the menus except for **INFO** mode that are also accessible after inputting the password.

The service password has higher priority than the user password.

4 identifier lines are printed out on the record; e.g., name of the customer's company.

Three adjustment procedures are repeated automatically at the set times selected by the user.

These adjustment times are in addition to the adjustment criteria of the isoCAL function!

These adjustment times are set by inputting hh.mm or hh (without the minutes).

The settings can be input without a "0" in front (e.g. 7.30, 12.00, 18.15).

Note:

"S" is displayed if the service mode was activated before the **Input-mode**.

MENU - Balance Operating Menu

SETUP	SELECTION
Config =>	Printout configuration
App =>	Application menu
Info =>	Balance/scale parameters
Menu =>	Balance/scale menu
Inout =>	User data
<< Config	App Info Menu Inout

La-002e.BMP

F2

SETUP	MENU	[	]
1	Balance/scale functions		
5	Interface		
6	Print in weighing mode		
8	Extra functions		
9	Reset menu		
<< App		v	>

La-013e.BMP

BAL.FUNC.	ADAPT FILT.	[	1-	1-	]
1	Minimum vibration				
2	Normal vibration				
3	Strong vibration				
4	Extreme vibration				
<< App		<	^	v	↓

La-014e.BMP

Max 5200g	d= 0.01g
0%	100%
0.00 g	
Cal	

La-015e.BMP

- Press **F2** while running the SETUP program.
- The symbol to the left appears.
- Use the keys **F1 - F4** (cursor function) to select the desired menu level and code digits.
- Press **F1** to set a new menu item.
- Press **SETUP** or **F6** to store new code settings and exit the operating menu.
- The balance returns to the weighing mode.

Note:

See the section entitled "Balance Operating Menu Settings **Master^{pro} LA**" starting on page 47 for an overview of the balance operating menu.

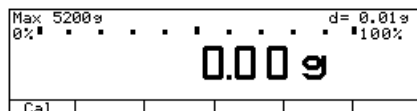
Activating the Service Mode

The LP/LA balances are equipped with a service mode that enables expanded calibration/adjustment functions.

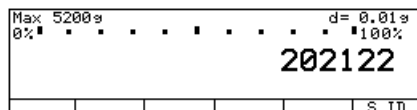
The service mode only remains active until:

1. you press **SETUP** to exit the menu, or
2. **CF** is pressed when the "Service CAL Selection" menu is displayed

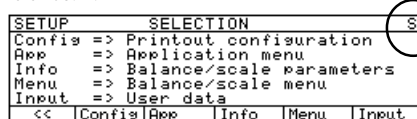
The weighing mode is active (0.00 g is displayed - depending on the balance model)



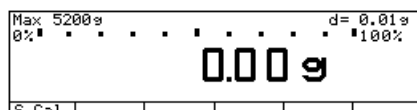
La-015e.BMP



La-016e.BMP



La-017e.BMP



F6
La-018e.BMP

- Enter the service password: »202 122« (this has a higher priority than a customer password) and
- Press **SETUP**.
- The SETUP menu is displayed and an "S" is shown in the lower left corner.
- The service mode is now active.
- Press **F6** to exit the SETUP menu; press **F6** again to activate the calibration/adjustment menu.

Calibration/Adjustment in the Service Mode

Adjustment options in the service mode:

- | | | |
|--------------------------|----|---|
| 1. int. adjust | *- | Internal span adjustment |
| 2. def. ext.lin.std.wt. | - | External linearization with default weight |
| 3. ext. linear. user wt. | - | External linearization (enter weight value in the INPUT mode) |
| 4. reprotect | *- | Reproducibility test of the internal adjustment |
| 5. det. int. wt. | - | Overwrite the internal calibration weight |
| 6. det. ext. wt. | - | Determine and store the external weight (user-defined weight) |
| 7. disp. *10 | - | 10-fold increased display resolution and SuperRange |
| 8. ext.adj.std.wt. | *- | External span adjustment with stored weight (entered in the INPUT mode) |

Note:

In most cases it is not necessary to use service software programs (PSION or Sartocas) to adjust the LP/LA balances, because all necessary adjustment functions are accessible when the balance is in the service mode.

You only need to use the adjustment software if the current adjustment error is greater than 2% or if multiple adjustment steps are necessary due to unfavorable ambient conditions.

* Can also be adjusted without activating the service mode

External Linearization

There are two ways to perform external linearization:

1. with the Sartorius service software (PSION Server or SARTOCAS Service Software for PCs) - if the adjustment error is greater than $\pm 2\%$, or
2. as described in this manual, with the balance in the service mode.

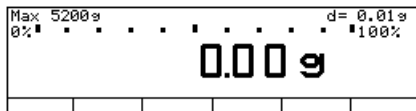
External Linearization with Manual Input of Weight Values

If adjustment weights of sufficient accuracy are not available (at least one digit greater than the display accuracy) or if no calibration certificate is available, you can program the balance with the weight value including all necessary decimal places.

If this is the case, follow the steps listed below:

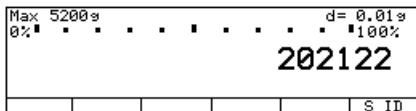
Manual Input of the Linearization Weights

- Press  to turn on the balance.
- The balance performs a self-test.
- 0.00 g is displayed (depending on balance model)



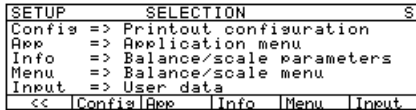
La-001e.BMP

- Activate the SERVICE mode:
- Enter the service password: >202 122<
- Press .



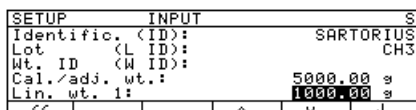
La-016e.BMP

- Press **F1** to activate the INPUT mode.



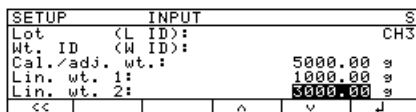
La-019e.BMP

- Press **F2** to activate the "LIN.WEIGHT 1" menu item and use the numeric keys to enter the calibration weight to input the adjustment weight value, including all necessary decimal points, for the first linearization weight in accordance with the calibration certificate.



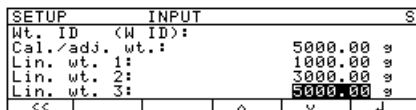
La-020e.BMP

- Press **F1** to store the weight value entered, and then press **F2** to proceed to the "LIN.WEIGHT 2" menu item.



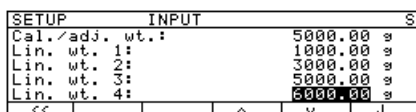
La-021e.BMP

- Use the numeric keys to enter the calibration weight for the second linearization weight, press **F1** to store, and then press **F2** to proceed to the "LIN.WEIGHT 3" menu item.



La-022e.BMP

- Use the numeric keys to enter the calibration weight for the third linearization weight, press **F1** to store, and then press **F2** to proceed to the "LIN.WEIGHT 4" menu item.



La-023e.BMP

- Use the numeric keys to enter the calibration weight for the fourth linearization weight and press **F1** to store.

- After defining all 4 linearization weights, press  to exit the INPUT mode.

- The balance returns to the weighing mode.

Performing External Linearization

After defining the linearization weights, the linearization procedure can be started.

- Activate the service mode again (see page 11).



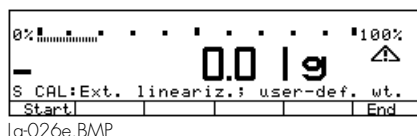
- Press **F6** to activate the adjustment menu.
- Press **F1** to activate the "EXT. LINEAR USER WT." menu item, or tare the balance and press **F6** to start the linearization routine.



- The first linearity weight is displayed.
Center the corresponding weight on the weighing pan.

Note:

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is performed automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last linearization is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).



- The deviation from the last adjustment is displayed.
- If the error exceeds the tolerance limits, either press **F6** to make the corresponding adjustment, or press **F1** to cancel the adjustment procedure.



- The second linearity weight is displayed.
- Center the corresponding weight on the weighing pan and press **F6** to confirm.



- Repeat this procedure until all linearity weights have been adjusted.
- After defining the last linearization weight, all adjustment weights must be removed from the weighing pan and the balance zeroed again.



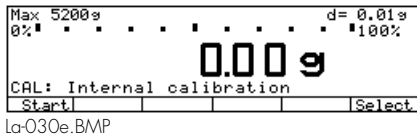
- After concluding the linearization routine, the service mode remains active until you press **[CF]**.

If linearization weights of sufficient accuracy are available (at least one digit greater than the display accuracy) it is not necessary to defined the linearization weights before performing the adjustment routine (in INPUT mode). In this case, linearization can be started immediately by selecting the menu item "Lin. extern std. wt.".

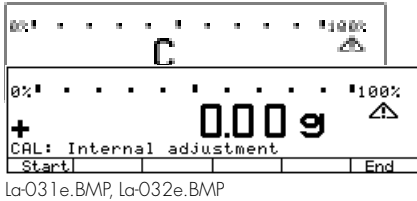
Perform the routine in accordance with the routine described above.

Internal Span Adjustment

All LP/LA balances are equipped with a built-in motorized calibration weight.



- Turn on the balance and allow it to warm up.
- Zero the balance, if necessary (press **TARE**).
- Press **F6**; press **F1** to select "INT. CALIBR." or "INT. ADJUST" and press **F6** to start the internal span adjustment.



- The built-in calibration weight is applied automatically, the weight is stored, and the deviation from the last span adjustment is displayed.

Note:

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is made automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last linearization is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).

- If the error exceeds the tolerance limits, press **F6** to make the corresponding adjustment, or press **F1** to cancel the adjustment procedure.
- The balance then returns to the weighing mode.



External Span Adjustment

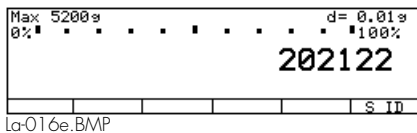
There are two ways to perform external span adjustment:

1. with the Sartorius service software (PSION Server or SARTOCAS Service Software for PCs) - if the adjustment error is greater than $\pm 2\%$, or
2. with the balance in the service mode, as described in this manual.

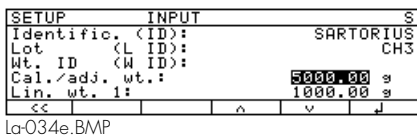
External Span Adjustment with Manual Input of Weight Values

You only need to input the calibration weight for span adjustment if:

1. the available weight is not sufficiently accurate (at least one digit greater than the display accuracy), or
2. you are working with a weight other than the recommended weight (e.g., with a heavier weight, close to the maximum capacity).

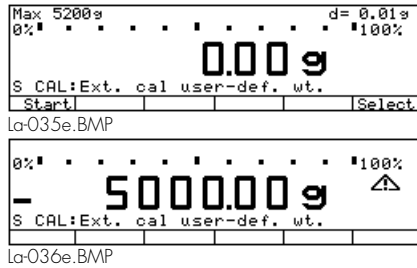


- Press **1/0** to turn on the balance.
- The balance performs a self-test.
- When 0.00 g is displayed (depending on balance model), enter the service password "202 122" and press **SETUP**.



- Press **F1** to activate the INPUT mode.
- Press **F2** to select the "CAL./ADJ. WEIGHT" menu item.
- Enter the conventional weight value, including all decimal places, using the numeric keys.
- Press **F1** to store the adjustment and press **F6** to exit the SETUP program.

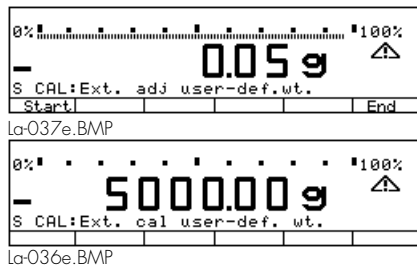
External Span Adjustment with Stored Weight



- Activate the service mode (see page 11).
- Press **F6** and then press **F1** to select the "CAL. EXT. ADJUST. WT." menu item.
- Tare the balance, if necessary, and then press **F6** to start external span adjustment.
- Center the prompted calibration weight on the weighing pan.

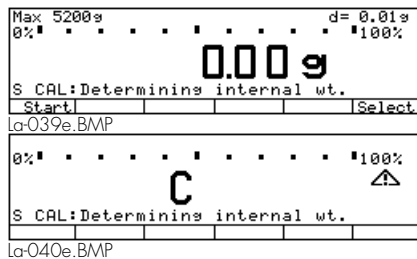
Note:

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is made automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last linearization is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).



- If the error exceeds the tolerance limits, press **F6** to make the corresponding adjustment, or press **F1** to cancel the adjustment procedure.
- The balance then returns to the SELECT mode.

Overwriting the Internal Calibration Weight



- Activate the Service mode (see page 11).
- Perform external span adjustment.
- Press **F1** to select the "DET. INT. WT." menu item.
- Tare the balance, if necessary, and press **F6** to overwrite the calibration weight.
- The built-in calibration weight is applied automatically and measured.

Note:

Depending on the menu code setting (under 1 10 X), either calibration/adjustment is made automatically (code 1 10 1: calibrate, then auto-adjust), or the deviation from the last linearization is displayed, but no adjustment is made until you press a key (code 1 10 2: calibrate, then manual adjust).



- If the error exceeds the tolerance limits, press **F6** to make the corresponding adjustment, or press **F1** to cancel the adjustment procedure.



- Press **[CF]** if you need to cancel this routine.
- The balance then returns to the SELECT mode.

Calibration with a User-defined Weight

You can define any weight as the calibration weight.

The weight value of this weight is stored automatically in the **INPUT** mode under the "CAL./ADJ.WT." menu item.



- Activate the service mode (see page 11).
- Perform external or internal span adjustment.
- Press **F6** to select the "DET.USER WT." menu item.

- Tare the balance if necessary, and then press **F6** to start the calibration routine.



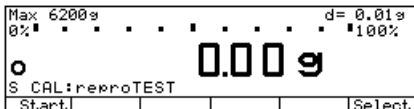
- The last calibration weight that was entered in the INPUT mode is displayed.



- Center the calibration weight you wish to define on the weighing pan.
- The deviation from the stored weight is displayed.
- The new user weight can be selected and stored by pressing **F6**.
- Press **F1** to cancel the routine.

- The balance returns to the SELECT mode.

reproTEST

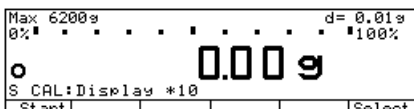


This tests the reproducibility of the automatic application of the built-in calibration weight.

The internal weight is applied, measured and removed 6 times.

At the end of the routine, the reproducibility value is displayed.

High Resolution - Display * 10



The balance switches to SuperRange resolution and an additional digit is displayed (for test purposes and initial verification).

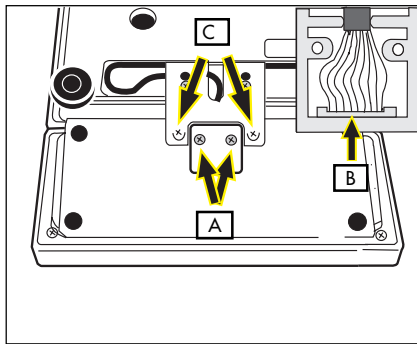
Servicing the Weighing Cell

Note:

Always disconnect the balance from AC power and then disconnect all cables from the balance before beginning service work!

LP balances are equipped with either a small single-casting block system, a brass system or a large single-casting block system.

Removing the Display and Control Unit from the Weighing Cell



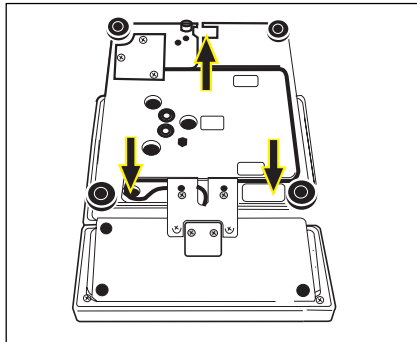
Unt0001.EPS

- Disconnect the balance from AC power.
- Remove all movable parts from the balance; e.g., weighing pan, pan flexures, draft shield, etc.
- Turn the balance over; place it on a soft surface to avoid damaging the weighing system.
- Remove the 2 screws (A) from the base pan.
- Disconnect the plug indicated on the left (B).
- Remove the fastening screws (C) from the display and control unit retainer and take the cable out of the strain relief clamp.
- Return the balance to the upright position and remove the dustcover, placing it to one side.

Opening the Weighing Cell (Small Single-Casting Block System, Brass System)

Note:

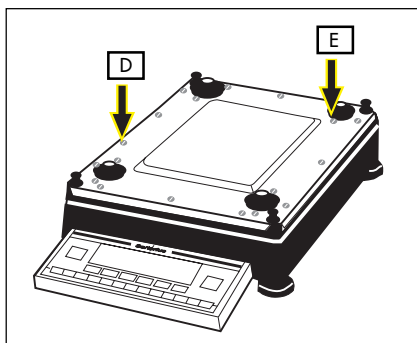
If you remove the seal (warranty sticker), the balance is no longer covered under the warranty.



Unt0002.EPS

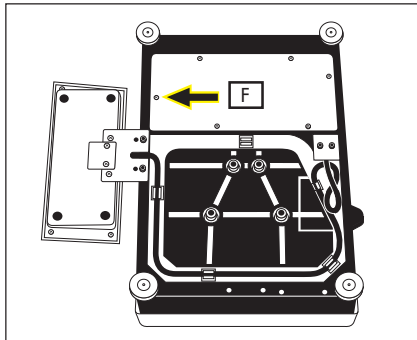
- Disconnect the balance from AC power.
- Remove all movable parts from the balance; e.g., weighing pan, pan flexures, draft shield, etc.
- Carefully place the balance on its side and remove the 3 screws from the base pan.
- Return the balance to the upright position and remove the dustcover, placing it to one side.
- You can now access the weighing system and the balance PCB.
- Follow the above steps in reverse order when closing the balance.

Opening the Weighing Cell (Large Single-Casting Block System)

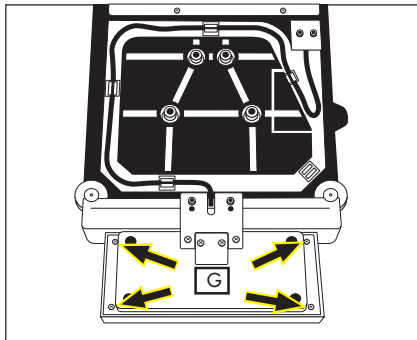


Grblo001.EPS

- Disconnect the balance from AC power.
- Remove the weighing pan.
- Remove the screws (E) that fasten the seals (103) and remove the 4 seals.
- Remove the 13 fastening screws (D) from the cover plate (104), remove the cover plate and place it to one side.



LC03.EPS



LC02a.EPS

Accessing the Main PCB

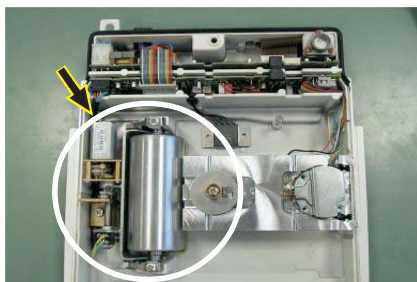
- Carefully tip the weighing platform on its side.
- Remove the 6 screws (F) from the bottom of the weighing platform; remove the cover plate (204) and place it to one side.
- The main PCB is now accessible.
- Follow the above steps in reverse order when closing the weighing platform.

Opening the Display and Control Unit

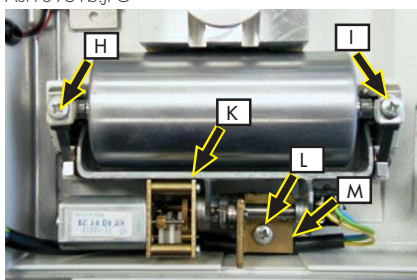
- Disconnect the balance from AC power.
- Remove all movable parts from the balance; e.g., weighing pan, pan flexures, draft shield, etc.
- Carefully place the balance on its side and remove the 4 screws (G) from the base pan of the display and control unit.
- Return the balance to the upright position, carefully lift the front panel of the display and control unit approx. 4 cm and then lay it down in front of the balance.
- The digital PCB and the display module are now accessible.
- Follow the above steps in reverse order when closing the display and control unit.

Replacing the Built-in Calibration Weight

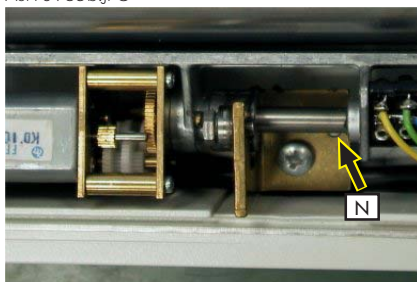
- Unplug the equipment from the AC adapter or power supply.
- Open the load cell (see Service Manual, page 27).
- The motorized calibration weight applicator and the built-in calibration weight are now visible.
- Loosen screws (H) and (I) four turns and rotate the flexures 180° (toward the rear).
- Press down on the weight lifting unit (K) to relieve strain and remove screw (L).
- Remove the brass angle bracket (M).
- Unplug the cable from the connector.
- Lift the motorized calibration weight applicator out of the load cell.
- You can now access the built-in calibration weight for replacement.



Aut10961b.JPG



Aut10965b.JPG



Aut10965b.JPG

Important:

Always wear gloves when handling the internal calibration weight.

Do not allow oil or dirt to come into contact with the weight.

- After positioning the new weight, install the weight lifting unit (K) and connect the cable.
Press the cable into the base of the housing to prevent it getting caught during reassembly.
- Install the brass angle bracket and (M) turn screw (L) approx. four times.
- Push the weight lifting unit (K) down gently, adjust the brass angle bracket (M) (center pin (N) in the hole) and tighten screw (L).
- Rotate the flexure under screw (H), and then the two flexures under screw (I), 180° to return them to the original position, and tighten the screws until they are in contact with the weight.
- Now re-adjust screw (H) ¼-turn in the other direction and press down on the weight lifting unit (K) to make sure the built-in weight is not held in place. Repeat this procedure with screw (I).

After replacing the built-in calibration weight, adjust the scale as follows:

1. Perform external calibration/adjustment;
2. overwrite the internal calibration weight value;
3. perform internal calibration/adjustment;
4. test the scale.

Note:

Calibration/Adjustment

Calibration/Adjustment Sequence

Check and, if necessary, adjust the following factors:

- Preload
- Reproducibility (standard deviation)
- Off-center load
- Span
- Linearity

Note:

The test procedures described in the following correspond to those described in the Sartorius Standard Operating Procedures, numbers WKD-037-01 and WKD-038-01. These also correspond to the specifications listed in the "Service Specifications Sheets for Master^{pro} Balances."

Prerequisites for Calibration/Adjustment

The following conditions are required for calibration/adjustment of the balance.

Ambient Conditions

- An area with favorable ambient conditions
- A level surface, free of vibration
- No direct source of heat radiation
- No drafts

Operating Conditions

- The balance must be leveled using the level indicator
- The balance must have warmed up for the required period of time

Preload

You only need to check the preload if:

- the weighing system has been rebuilt (i.e., if the linkage was replaced)
- the weighing system was damaged
- the user wishes to change the preload (e.g., to use a special weighing pan) while retaining the entire weighing range

Note: There are restrictions on the extent to which the preload can be changed! In general, an increase in the preload by more than 10% is not permissible. If necessary, contact the International Service Support or Customer Support in Goettingen.

There are 2 different ways to check/adjust the preload:

- With the service software (SARTOCAS program or PSION Server)
- With a digital voltmeter

Checking the Preload

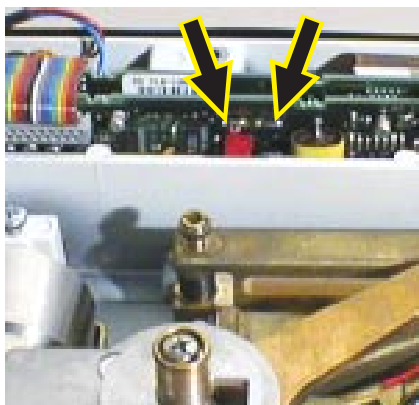
Note: If the preload values have changed significantly, this indicates a mechanical or electronic error.

1. Checking the Preload with the Service Software

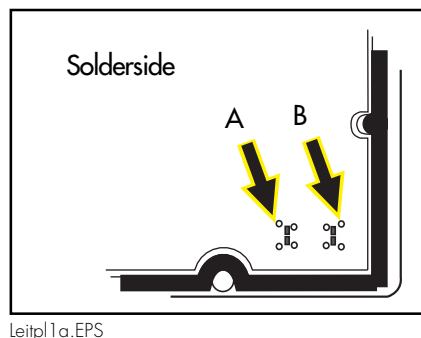
- Replace the pan flexures and weighing pan on the balance, connect the power supply and turn on the balance.
- Activate the BPI mode (see page 10).
- Connect the PSION Server or a PC/laptop with the SARTOCAS program.
- Start the service software and save the data records that are currently in the balance.
- Select the ELECTR.ZERO item from the DIAGNOSTICS submenu.
- Place the required preload test weight on the balance (see "Adjustment Specifications," page 31) and read the PSION server/PC display.
- Compare the value displayed to the tolerance range given in the adjustment specifications.
- If the displayed value exceeds the tolerance limits, the preload must be adjusted.

2. Adjusting the Preload with a Digital Voltmeter

- Solder 2 instrument leads to the balance PCB at positions A and B (parallel to the measuring resistor) and connect them to the digital voltmeter.
- Replace the pan flexures and weighing pan on the balance, connect the power supply and turn on the balance.
- Set the digital voltmeter for a range of 20 V DC and place the required preload on the balance (see "Adjustment Specifications," page 31).



Messwi02.BMP



- Compare the voltage displayed on the digital voltmeter with the preload voltage given in the „Adjustment Specifications.“ If necessary, switch the voltmeter to a lower measuring range (e.g., 200 mV DC).
- If the preload voltage measured exceeds the tolerance limits listed in the specifications, you must adjust the preload.
- The data for preload test weights and preload voltage are listed in the “Adjustment Specifications” below.

Adjustment Specifications

LA/LP balances with brass system:

LA/LP1200S, LA/LP3200D, LA/LP4200S, LA/LP5200P, LA/LP6200S

Models	Preload test weight	Preload voltage	A/D converter level control
LA/LP1200S	600 g	–400 mV to +400 mV	20 % ± 2.3 %
LA/LP3200D	1500 g	–260 mV to +260 mV	20 % ± 1.5 %
LA/LP4200S	300 g	–270 mV to +270 mV	20 % ± 1.6 %
LA/LP5200P	300 g	–270 mV to +270 mV	20 % ± 1.6 %
LA/LP6200S	300 g	–270 mV to +270 mV	20 % ± 1.6 %

LA/LP balances with small single-casting block system:

LA/LP420, LA/LP220S, LA/LP620S, LA/LP620P, LA/LP820, LA/LP2200, LA/LP2200S, LA/LP2200P, LA/LP4200, LA/LP6200, LA/LP12000S, LA/LP12000P

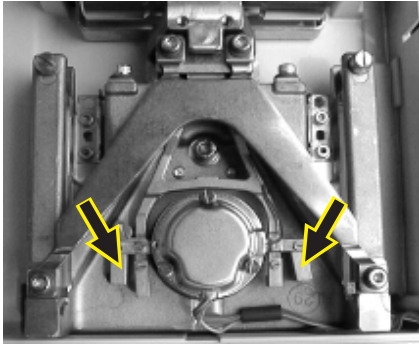
Models	Preload test weight	Preload voltage	A/D converter level control
LA/LP220S	300 g	–360 mV to +360 mV	20 % ± 1.2 %
LA/LP420S	2150 g	–590 mV to +590 mV	20 % ± 3.4 %
LA/LP620S	300 g	–360 mV to +360 mV	20 % ± 2.1 %
LA/LP620P	300 g	–360 mV to +360 mV	20 % ± 2.1 %
LA/LP820	300 g	–360 mV to +360 mV	20 % ± 3.4 %
LA/LP2200	2150 g	–590 mV to +590 mV	20 % ± 3.4 %
LA/LP2200S	2150 g	–590 mV to +590 mV	20 % ± 3.4 %
LA/LP2200P	2150 g	–590 mV to +590 mV	20 % ± 3.4 %
LA/LP4200	6000 g	–200 mV to +200 mV	20 % ± 1.2 %
LA/LP6200	6000 g	–200 mV to +200 mV	20 % ± 1.2 %
LA/LP12000S	3000 g	–200 mV to +200 mV	20 % ± 1.2 %
LA/LP12000P	6000 g	–200 mV to +200 mV	20 % ± 1.2 %

LA/LP balances with large single-casting block system:

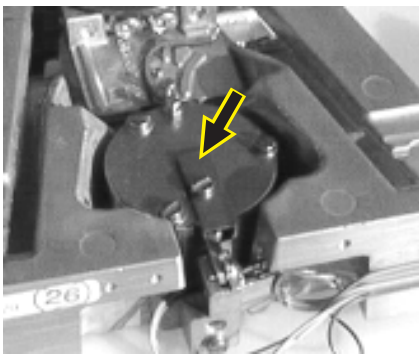
LA/LP16000S, LA/LP34000P, LA/LP34

Models	Preload test weight	Preload voltage	A/D converter level control
LA/LP16000S	17 kg	–130 mV to +130 mV	20 % ± 0.75 %
LA/LP34000P	17 kg	–130 mV to +130 mV	20 % ± 0.75 %
LA/LP34	17 kg	–130 mV to +130 mV	20 % ± 0.75 %

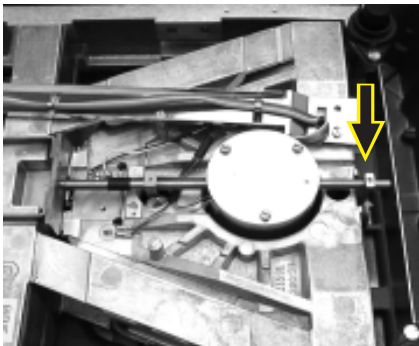
Adjusting the Preload



Messy05b.TIF



Blosy01a.TIF



System1.TIF

- Preparations for adjusting the preload are the same as for checking the preload.
- The preload is changed as follows:
 - in Master^{PRO} balances with the brass system, preload weights are removed from or added to the end of the lever.
 - in Master^{PRO} balances with the small single-casting block system, preload weights (arrow) are removed from or added to the lever over the magnet.
 - in Master^{PRO} balances with the large single-casting block system, preload weights are removed or added (rough adjustment) or position of the preload weight on the lever is moved (fine adjustment) - to do this, you will need to loosen the set screw.
- The balance remains loaded with the preload test weight during this procedure.
- Check the display of the PSION server/PC or the digital voltmeter after each change.
- If the preload voltage/preload weight value is within the tolerance range, tighten down the set screw.

Note:

After you adjust the preload, the off-center load, linearity and span must be checked and, if necessary, adjusted.

Reproducibility

- The required test weight is listed in the "Service Specifications Sheet for Master^{pro} Balances."
- Unload the weighing pan and press **TARE** to tare.
- Center the test weight on the weighing pan.
- Write down the value displayed.
- Repeat this procedure 5 times.
- The reproducibility value is derived from these 6 values as follows:

$$\frac{(\text{highest disp. value} - \text{lowest display value})}{3}$$

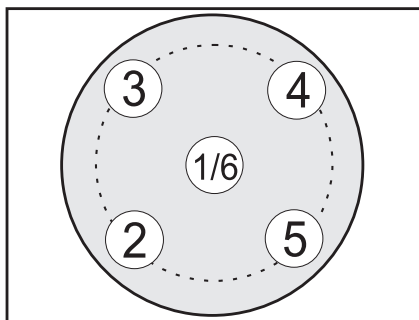
- Compare the resulting value to the tolerance range given.
- If the calculated value exceeds the tolerance limits, this may be caused by one of the following:
 - Foreign matter/dirt in the weighing system
 - Mechanical parts touching the weighing system (pin in the magnet cover)
 - Bent/defective bending beam

Off-center Load

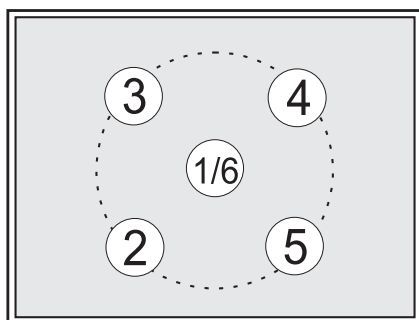
Checking the Off-center Load

Note:

The specifications for test weights and tolerances are listed in the "Service Specifications Sheets for Master^{pro} Balances" (see page 46).



Ecklr02.EPS



Eckle01.EPS

- Place the test weight in position 1 on the weighing pan and press **TARE** to tare the balance.
- Place the test weight on positions 2, 3, 4, 5 and 6 in sequence and write down the values, with plus or minus sign, displayed at stability.
- Master^{pro} Balances with round weighing pan
- Master^{pro} Balances with rectangular weighing pan
- Compare the values with the tolerance limits listed in the "Service Specifications for Master^{pro} Balances" (see page 46).
- If the values exceed the tolerances, the off-center load must be adjusted.

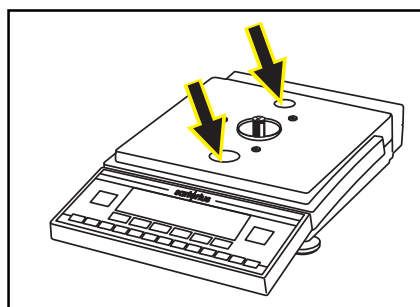
Adjusting the Off-center Load

Master^{pro} Balances with the Small Single-Casting Block System

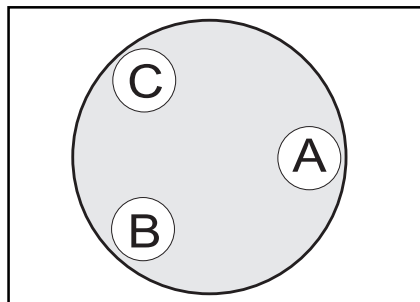
Adjusting the off-center load requires specialized knowledge of small single-casting block systems (especially for adjusting balances with a readability of 1 mg!) which cannot be described in detail in this service manual. However, you can acquire the required knowledge by attending the corresponding service training course on this topic. For more information, please contact Sartorius AG in Goettingen, Germany, or the Sartorius Training Center.

Master^{pro} Balances with the Brass System

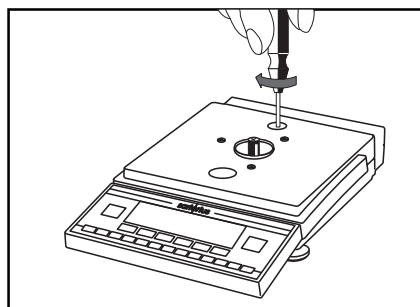
3-Point Adjustment (rough)



LP-ob01.EPS



Ecklr01.EPS



LP-ob01.EPS

- The off-center load is adjusted by turning the 2 adjustment screws as described in the following.
- The adjustment screws are accessed through 2 openings in the upper part of the balance housing.
- To access these openings, remove the weighing pan and then remove the 2 stoppers from the upper part of the housing (arrows).

- Place the test weight in position A on the weighing pan and press **TARE** to tare the balance.
- Place the test weight on positions B and C in sequence and write down the values, with plus or minus sign, displayed at stability.

- Example:

Pos. A	Press TARE	0.00 g
Pos. B		+0.05 g
Pos. C		–0.04 g

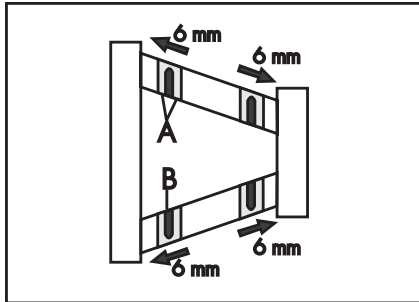
- Only adjust the side with the greater error (absolute value) as follows:
Negative error value:
Turn the screw counter-clockwise
Positive error value:
Turn the screw clockwise
- Change the screw position in small increments and re-measure the off-center load in positions A through C after each adjustment.
- Repeat this procedure until the off-center load error is within the tolerances at all 3 positions.
- Then re-measure the off-center load in 5 positions (see page 33).
- If the off-center load error exceeds the tolerances when you perform this test, perform the 5-point fine adjustment.

Master^{pro} Balances with the Large Single-Casting Block System

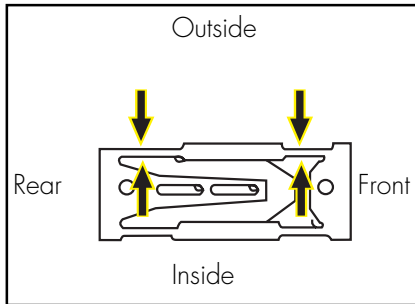
Note:

This adjustment should only be performed by trained, qualified technicians. There is a risk of doing irreparable damage to the weighing system during this adjustment procedure.

The adjustment is performed by filing down the upper guide of the load cell.



ICeck-0.EPS



Block-01.EPS

- Remove the cover plate (104) and re-check the off-center load (the values may differ from those measured with the plate in place).
- Apply the file only to the upper or lower side of the thin parts on the upper guide, 6 mm behind the per-milled part.
- Use a round file with a diameter of 2 - 3 mm.
- Hold the file at an angle of approx. 2°.

- Refer to the displayed values and the following overview to determine the exact spot that needs to be filed.
- When the displayed values are all within the tolerance ranges, close the balance.

Legend:	RTRI	=	Rear, Top, Right, Inside
	RTRO	=	Rear, Top, Right, Outside
	RTLI	=	Rear, Top, Left, Inside
	RTLO	=	Rear, Top, Left, Outside
	FTRI	=	Front, Top, Right, Inside
	FTRO	=	Front, Top, Right, Outside
	FTLI	=	Front, Top, Left, Inside
	FTLO	=	Front, Top, Left, Outside

Note:

Inside = Lower side of the upper guide
Outside = Upper side of the upper guide

Overview for filing:

1 	2 	3 	4
5 	6 	7 	8

ICeck-1.TIF

Span Accuracy

Checking the Span Accuracy

Note:

You must use certified calibrated or verified weights to check the span accuracy.

It must be established that the weights are traceable to a national standard. The calibration or verification certificate must indicate the nominal mass value, the deviation from the nominal value, and the uncertainty of measurement. The exact value (**conventional mass value**) for each weight is calculated from the nominal value and the deviation from the nominal value.

These requirements do not apply if the accuracy of the weight used for checking the span accuracy is at least one digit greater than the display accuracy.

- See (page 48) the “Service Specifications Sheets for Master^{pro} Balances” for the required test weight.
- Unload the balance and press **TARE** to tare
- Perform internal span adjustment (LP see page 14) / (LA see page 24) .
- After performing span adjustment, center the test weight on the weighing pan.
- Read off the weight at stability and compare it to the tolerance limits listed.

- If the resulting value exceeds the tolerance limits, you need to perform external span adjustment and then overwrite the internal calibration weight (LP see page 15 / LA see page 25, “Overwriting the Internal Calibration Weight”).

Linearity

Checking the Linearity

There are 2 different procedures for checking the linearity, depending on the balance specifications:

- for balances with a resolution of less than 1,000,000 digits
- for balances with a resolution of more than 1,000,000 digits and for all PolyRange balances

Note:

Before checking the linearity, the span adjustment must be checked and, if necessary, adjusted.

Checking the Linearity on Balances with Less Than 1,000,000 Digits

- See the “Service Specifications Sheets for Master^{pro} Balances” for the required test weight.
- Check the linearity for the entire weighing capacity: at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ and maximum load.
- Two additional measurements are made for checking DualRange balances.
- Unload the balance and press **TARE** to tare.
- Center the first test weight on the weighing pan and write down the weight displayed at stability (L1).
- Unload the balance and press **TARE** to tare.
- Center the second test weight on the weighing pan and write down the weight displayed at stability (L2).
- Unload the balance and press **TARE** to tare.
- Center the third test weight on the weighing pan and write down the weight displayed at stability (L3).
- Unload the balance and press **TARE** to tare.
- Center the fourth test weight on the weighing pan and write down the weight displayed at stability (L4).
- For DualRange balances, perform 2 additional measurements as described above (L5, L6).
- Calculate the linearity error for each point tested:
Error = displayed value – value of the test weight.
- The L1, L2 and L5 test points must not exceed the tolerance limits.
- The L3, L4 and L6 test points can have up to twice the permissible error.
- If the resulting linearity errors exceed the tolerance limits, the linearity must be re-adjusted (see page 13, “External Linearization”).

Checking the Linearity on Balances with More Than 1,000,000 Digits and PolyRange Balances

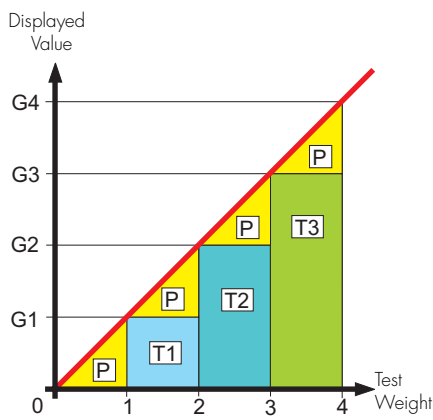
Note:

For balances with a resolution of more than 1,000,000 digits (e.g., the LP1 200S) and for PolyRange balances, the linearity is checked using the relative method.

4 measurements are taken.

For the first measurement, the test weight is placed on the balance without a tare load.

For the following measurements, the test weight is measured 3 times, with 3 different tare weights.



- See the “Service Specifications Sheets for Master^{pro} Balances” for the required test weights and tare weights.

- Unload the balance and tare it.
- Center the test weight on the weighing pan.
- Write down the value displayed at stability (G1).
- Unload the balance.

- Place the first tare weight on the weighing pan and tare the balance.
- Center the test weight on the weighing pan.
- Write down the value displayed at stability (G2).
- Unload the balance.

- Place the second tare weight on the weighing pan and tare the balance.
- Center the test weight on the weighing pan.
- Write down the value displayed at stability (G3).
- Unload the balance.

- Place the third tare weight on the weighing pan and tare the balance.
- Center the test weight on the weighing pan.
- Write down the value displayed at stability (G4).
- Unload the balance.

- The linearity error is calculated according to the following formula:

$$\text{linearity Error L} = \frac{\text{Highest disp. value} - \text{lowest disp. value}}{2}$$

- If the resulting linearity error exceeds the tolerance limits, the linearity must be re-adjusted (LP see page 13 / LA see page 23, “External Linearization”).

Repairing the Balance

Repairing the Weighing System

Do not attempt to repair the weighing system; send defective equipment to a Sartorius Repair Center, or to the main factory in Goettingen. Temperature compensation may be required; this can only be performed in Goettingen.

Repairing the Electronics

Note:

Do not attempt to repair electronic components in Master^{pro} balances.

In case of defects, replace the entire PCB.

To replace the 2nd main PCB, you must use the PSION Server or SARTOCAS software program. Make sure you order/install a PCB that has been programmed for the correct balance **model and serial number**.

Do not install a PCB from a different balance, because every data record is specific to the weighing system in the balance.

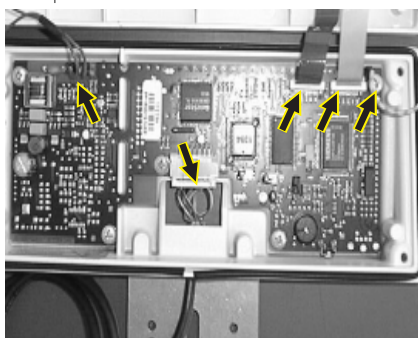
Replacing the PCB in the Display and Control Unit

Before replacing the PCB, make sure you either:

- write down the user's settings (entered in the INPUT mode and/or in application programs), or
- copy the settings from the old to the new PCB.
- Disconnect the balance from AC power.
- Open the display and control unit (see page 28 for instructions).

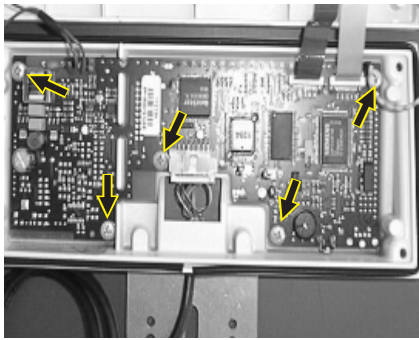


Anzkop-2.TIF



Anzkop1a.TIF

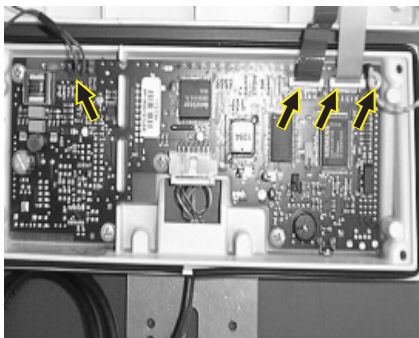
- Disconnect the wires to the keypad overlay, the display, the ground connection and the display backlighting and remove the entire front panel and place it to one side.
- Disconnect the weighing cell cable.
- Remove the 5 fastening screws and remove the defective PCB.
- If possible (if it is not defective), copy the operating menu from the old PCB to the new one. To do this:
 - Carefully remove the operating menu EEPROM from its socket.
 - Plug it into the corresponding socket on the new PCB.



Anzkop1a.TIF

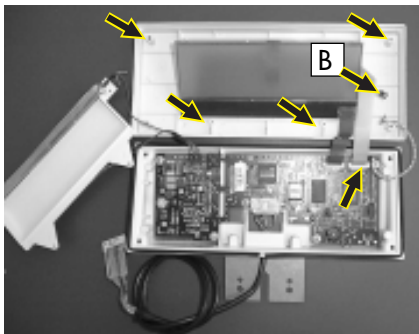
- Fix the new PCB in place with the 5 fastening screws.
- Reconnect all the connecting cables and plugs.
- Position the seals.
- Position the front panel and fasten it with the 4 screws.
- Connect the balance to AC power and turn it on.
- The error code "ERR340" is displayed.
- Turn the balance off and back on again.
- If the operating menu EEPROM was not moved from the old to the new PCB, the user's settings will need to be input now (via the INPUT mode and the MENU).

Replacing the Display and Control Unit, Backlighting Module, Keypad Overlay



Anzkop1a.TIF

- Disconnect the balance from AC power.
- Open the display and control unit (see page 28 for instructions).
- Disconnect the wires to the keypad overlay, the display, the ground connection and the display backlighting.
- The front panel is now accessible
- Remove the 5 tin screws and remove the backlighting module.
- You can now remove and replace the display unit.
- Remove the adhesive keypad overlay from the front panel, remove any traces of adhesive remaining on the panel, and affix the new keypad overlay.
- Re-connect the ground connection to screw (B), position the display and the backlighting module, and fasten them with the 4 tin screws.
- Reconnect all the connecting cables and plugs.
- You can now close the display and control unit (see page 27).



Anzkop1.TIF

Replacing the Balance PCB

Note:

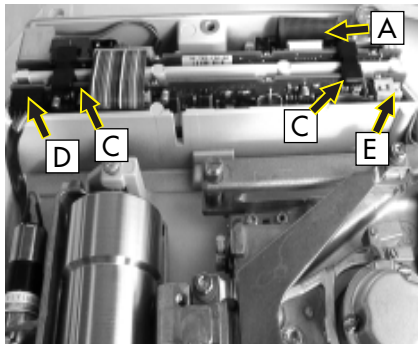
To replace the balance PCB, you need the SARTOCAS Service Program for PCs, version 1.10 or newer, or the PSION server, CAS version 4.5 or newer.

Before replacing the PCB, you should attempt to copy the data from the old PCB and store it.

This is done using the SARTOCAS Service Program for PCs, version 1.10 or newer, or the PSION server, CAS version 4.5 or newer.

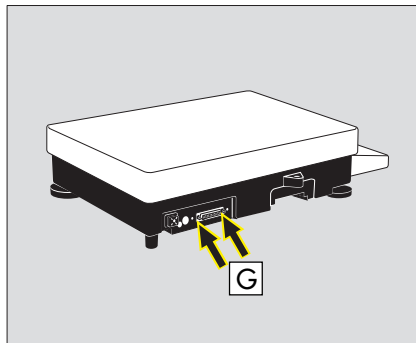
The required procedures are described in the corresponding program instructions.

If it is not possible to read the data records or if no replacement PCB is available, you can order a pre-programmed PCB from the Sartorius factory. Make sure you include the correct balance model and serial number when ordering the PCB.

Master^{pro} Balances with the Small Single-Casting Block System or Brass System

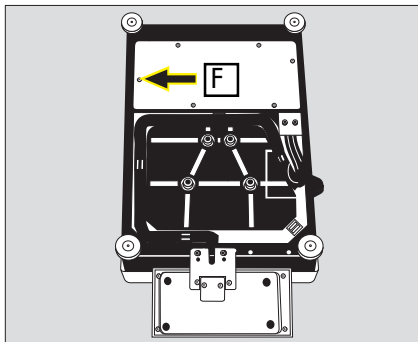
Platau-1.TIF

- Disconnect the balance from AC power.
- Open the weighing cell (see page 27).
- Disconnect the cables to the data output port (A) and the power supply.
- Remove both spacers (C) and carefully pull the PCB out of the base pan.
- Disconnect the cables to the weight application mechanism (D) and the weighing system (E).
- Follow these steps in reverse order to install the new PCB.
- Close the weighing cell (see page 27).
- The balance must now be programmed (if the PCB installed is not pre-programmed).
- After programming the balance, check the linearity and span.

Master^{pro} Balances with the Large Single-Casting Block System

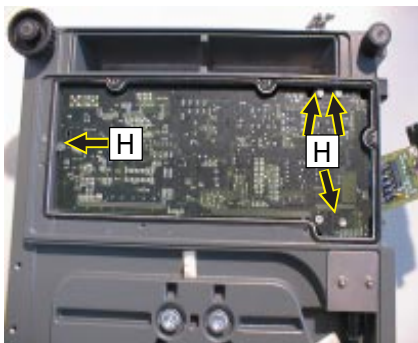
LPO8.EPS

- Disconnect the balance from AC power.
- Remove weighing pan.
- Open weighing cell (see page 27).
- Remove the fastening screws (G) and take out the data output port.



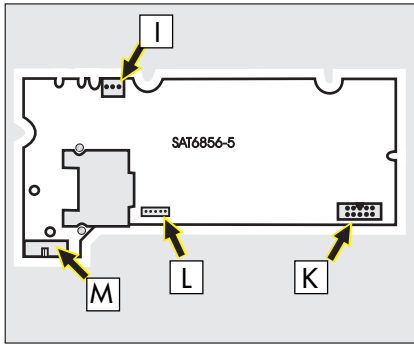
LPO3a.EPS

- Carefully turn the balance over.
- Remove the 6 screws (F) from the cover plate.



PIC00001.JPG

- Remove the 4 screws (H) from the main PCB.



SAT6856.EPS

Note:

- Disconnect the following cables:
Power supply (I); weighing system (K);
weight application mechanism (L); display and control unit (M).
- You can now remove the main PCB.
- Follow these steps in reverse order to install the new PCB.
- The balance must now be programmed (if the PCB installed is not pre-programmed).

After programming the balance, check the linearity and span.

Pin Assignments

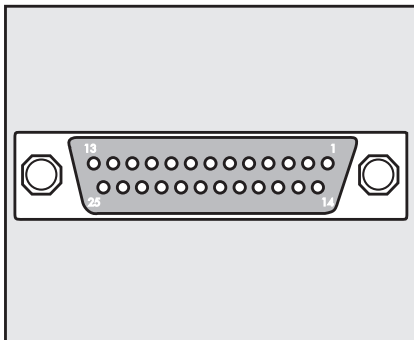
Female Interface Connector:

25 position D-Submini, DB25S, with screw lock hardware for cable gland

Note:

Warning When Using Pre-wired RS-232 Connecting Cables:

RS-232 cables purchased from other manufacturers often have incorrect pin assignments for use with Sartorius balances. Be sure to check the pin assignments against the chart on the right before connecting the cable, and disconnect any lines marked "Internally Connected" (e.g., Pin 6). Failure to do so may damage or even or even completely ruin your balance and/or peripheral device.



AMP25pv.EPS

Pin Assignments:

Pin 1:	Signal Ground
Pin 2:	Data Output (TxD)
Pin 3:	Data Input (RxD)
Pin 4:	Signal Return (TxD/RxD)
Pin 5:	Clear to Send (CTS)
Pin 6:	Internally Connected
Pin 7:	Internal Ground
Pin 8:	Internal Ground
Pin 9:	Reset_IN**)
Pin 10:	- 12V
Pin 11:	+ 12V
Pin 12:	Reset_Out**)
Pin 13:	+ 5V
Pin 14:	Internal Ground
Pin 15:	
Pin 16:	
Pin 17:	
Pin 18:	
Pin 19:	
Pin 20:	Data Terminal Ready (DTR)
Pin 21:	Supply Voltage Ground "COM"
Pin 22:	Not Connected
Pin 23:	Not Connected
Pin 24:	Supply Voltage Input +15...25V
Pin 25:	+ 5V

*) = See "Additional Functions" in the Installation and Operating Instructions for information on changing the pin assignments

***) = Hardware restart

Balance Operating Menu Codes Master^{pro} LP

1	Balance Functions		
1 1	Adapt Filter		
1 1 1	Minimum vibration		
1 1 2 •	Normal vibration		
1 1 3	Strong vibration		
1 1 4	Extreme vibration		
1 2	Application Filter		
1 2 1 •	Final readout		
1 2 2	Filling mode		
1 3	Stability Range		
1 3 1	¼ digit		
1 3 2	½ digit		
1 3 3	1 digit		
1 3 4 •	2 digits		
1 3 5	4 digits		
1 3 6	8 digits		
1 5	Taring *		
1 5 1	Without stability		
1 5 2 •	With stability		
1 6	Auto Zero		
1 6 1 •	On		
1 6 2	Off		
1 7	Weight Unit 1		
1 7 2 •	Grams	g	
1 7 3	Kilograms	kg	
1 7 4	Carats *	ct	
1 7 5	Pounds *	lb	
1 7 6	Ounces *	oz	
1 7 7	Troy ounces *	ozt	
1 7 8	Hong Kong tael *	tl	
1 7 9	Singapore tael *	tl	
1 7 10	Taiwanese tael *	tl	
1 7 11	Grain *	GN	
1 7 12	Pennyweights *	dwt	
1 7 13	Milligrams	mg	
1 7 14	Parts per pound *	o	
1 7 15	Chinese tael *	tl	
1 7 16	Mommes *	m	
1 7 17	Austrian carats *	k	
1 7 18	Tola *	t	
1 7 19	Baht *	b	
1 7 20	Mesghal *	m	
1 8	Display Accuracy 1		
1 8 1 •	All digits		
1 8 2	Few digits for weight change		
1 8 5	Last digit off *		

1 9	CAL Key Functions		
1 9 1	External calibration; factory-def. wt. **		
1 9 3	External calibration; user-def. wt. **		
1 9 4	Internal cal./adjustment		
1 9 10	CAL key blocked		
1 9 11	reproTEST		
1 9 12 •	Selection mode		

1 10	Cal./Adjustment Sequence		
1 10 1 •	Calibrate, then auto adjust		
1 10 2	Calibrate, then manual adjust		

1 13	Tare/Zero with Power On *		
1 13 1 •	On		
1 13 2	Off		

1 15	isoCAL Function		
1 15 1	Off		
1 15 2	Only adjustment prompt		
1 15 3	On and reset application		
1 15 5 •	On without resetting app.		

2	Application Programs		
2 1	Program Selection		
2 1 1 •	Basic weighing function		
2 1 2	Toggle weight units		
2 1 4	Counting		
2 1 5	Weighing in percent		
2 1 6	Net-total		
2 1 7	Animal weighing		
2 1 8	Checkweighing		

3	Applications Parameters		
3 1	Weight Unit 2		
3 1 2	Grams	g	
3 1 3	Kilograms	kg	
3 1 4	Carats *	ct	
3 1 5	Pounds *	lb	
3 1 6	Ounces *	oz	
3 1 7	Troy ounces *	ozt	
3 1 8	Hong Kong tael *	tl	
3 1 9	Singapore tael *	tl	
3 1 10	Taiwanese tael *	tl	
3 1 11	Grain *	GN	
3 1 12	Pennyweights *	dwt	
3 1 13	Milligrams	mg	
3 1 14	Parts per pound *	o	
3 1 15	Chinese tael *	tl	
3 1 16	Mommes *	m	
3 1 17	Austrian carats *	k	
3 1 18	Tola *	t	
3 1 19	Baht *	b	
3 1 20	Mesghal *	m	

- = Factory setting
- * = Not available on verified balances
- ** = Only on verified balances

3 5			Weight Storage Mode
3	5	2 •	Display accuracy
3	5	3	Display acc. x 10

3 6			Dec. Places for Readout in Percent
3	6	1	None
3	6	2	1 decimal place
3	6	3 •	2 decimal places
3	6	4	3 decimal places

3 7			Animal Activity
3	7	1	Calm
3	7	2 •	Normal
3	7	3	Active

3 8			Start
3	8	1	Manual
3	8	2 •	Automatic

3 9			Printout
3	9	1	None
3	9	2 •	Average weight only
3	9	3	Average and calculated value

3 10			Autostart
3	10	1	On
3	10	2 •	Off

4 Checkweighing

4 2			Automatic Printout
4	2	1	On in OK range
4	2	2 •	Off

4 3			Activation of Port Lines
4	3	1 •	Within control range
4	3	2	Always on
4	3	3	Stability and checkweighing range
4	3	4	At stability
4	3	5	Stability + Checkwgh. range -> on

5 Interface

5 1			Baud Rate
5	1	1	150 baud
5	1	2	300 baud
5	1	3	600 baud
5	1	4 •	1,200 baud
5	1	5	2,400 baud
5	1	6	4,800 baud
5	1	7	9,600 baud
5	1	8	19,200 baud

5 2			Parity
5	2	2	Space *
5	2	3 •	Odd
5	2	4	Even

5 3			Number of Stop Bits
5	3	1 •	1 stop bit
5	3	2	2 stop bits

5 4			Handshake Mode
5	4	1	Software handshake
5	4	3 •	Hardware handshake after 1 char.

6 Print in Weighing Mode

6 1			Manual/Auto Print Mode
6	1	1	Manual without stability
6	1	2 •	Manual with stability
6	1	4	Automatic without stability
6	1	5	Automatic at stability
6	1	6	Automatic after weight change ***

6 2			Stop Auto Print
6	2	1	Use print key
6	2	2 •	Not possible

6 3			Time-dependent Auto Print
6	3	1 •	1 display update
6	3	2	2 display updates
6	3	4	10 display updates *
6	3	7	100 display updates *

6 4			Print on Request then Tare
6	4	1 •	Off
6	4	2	On

7 Print in Application Mode

7 1			Printing Application Parameters
7	1	1 •	Off
7	1	2	On, all parameters
7	1	3	On, only main parameters

7 2			Line Format
7	2	1	For raw data
7	2	2 •	For other apps./GLP

7 3			Auto Print for Net-Total
7	3	1 •	Last net value
7	3	2	Tare value

- = Factory setting
- * = Not available on verified balances
- *** = Auto print when the load change is >50d and stability is set to < 25d

8 Extra Functions**8 1 Menu Mode**

8 1	1 •	Change parameters
8 1	2	Read only

8 2 Acoustic Signal

8 2	1 •	On
8 2	2	Off

8 3 Keypad

8 3	1 •	Accessible
8 3	2	Blocked
8 3	3	Alphabetic keys blocks

8 4 External Switch Function

8 4	1 •	Print key
8 4	2	Tare key
8 4	3	CAL/isoTEST key
8 4	4	F function key
8 4	5	CF key
8 4	6	Toggle key

8 5 Display Backlighting

8 5	1 •	On
8 5	2	Off
8 5	3	Auto off after 4 minutes

8 6 Power-on mode

8 6	1 •	Off > on / standby
8 6	2	Off > on ¹⁾
8 6	4	Auto on

8 7 Automatic shutoff

8 7	1	After 4 minutes
8 7	2 •	Off

8 8 Control Port Function

8 8	1	Input
8 8	2 •	Output

8 10 ISO/GLP/GMP Printout

8 10	1 •	Off
8 10	2	Only for calibration/adjustment
8 10	3	Always on

9 Reset Menu**9 1 Restore Factory Settings**

9 1	1	Restore factory settings
9 1	2 •	Do not restore factory settings
9 1	10 ²⁾	Reset all parameters (incl. INPUT data)
9 1	11 ²⁾	Restore Laboratory version
9 1	12 ²⁾	Restore Industry version

• = Factory setting

²⁾ = Only accessible in the service mode

Balance Operating Menu Codes

Master^{pro} LA

1 Balance Functions

1 1 Adapt Filter

1	1	1	Minimum vibration
1	1	2 •	Normal vibration
1	1	3	Strong vibration
1	1	4	Extreme vibration

1 2 Application Filter

1	2	1 •	Final readout
1	2	2	Filling mode

1 3 Stability Range

1	3	1	¼ digit
1	3	2	½ digit
1	3	3	1 digit
1	3	4 •	2 digits
1	3	5	4 digits
1	3	6	8 digits

1 5 Taring *

1	5	1	Without stability
1	5	2 •	With stability

1 6 Auto Zero

1	6	1 •	On
1	6	2	Off

1 7 Weight Unit 1

1	7	2 •	Grams	g
1	7	3	Kilograms	kg
1	7	4	Carats *	ct
1	7	5	Pounds *	lb
1	7	6	Ounces *	oz
1	7	7	Troy ounces *	ozt
1	7	8	Hong Kong tael *	tl
1	7	9	Singapore tael *	tl
1	7	10	Taiwanese tael *	tl
1	7	11	Grain *	GN
1	7	12	Pennyweights *	dwt
1	7	13	Milligrams	mg
1	7	14	Parts per pound *	o
1	7	15	Chinese tael *	tl
1	7	16	Mommes *	m
1	7	17	Austrian carats *	k
1	7	18	Tola *	t
1	7	19	Baht *	b
1	7	20	Mesghal *	m

1 8 Display Accuracy 1

1	8	1 •	All digits
1	8	2	Few digits for weight change
1	8	5	Last digit off *

1 9 CAL Key Functions

1	9	1	External calibration; factory-def. wt. **
1	9	3	External calibration; user-def. wt. **
1	9	4	Internal cal./adjustment
1	9	10	CAL key blocked
1	9	11	reproTEST
1	9	12 •	Selection mode

1 10 Cal./Adjustment Sequence

1	10	1 •	Calibrate, then auto adjust
1	10	2	Calibrate, then manual adjust

1 13 Tare/Zero with Power On *

1	13	1 •	On
1	13	2	Off

1 15 isoCAL Function

1	15	1	Off
1	15	2	Only adjustment prompt
1	15	3	On and reset application
1	15	5 •	On without resetting app.

1 16 Start automatic adjustment

1	16	1	User-def. adjustment times
1	16	2 •	Bal./ scale adjustment interval

1 17 GLP/GMP-compliant printout

1	17	1 •	Automatic if GLP is selected
1	17	2	On request, from data record

5 Interface

5 1 Baud Rate

5	1	1	150 baud
5	1	2	300 baud
5	1	3	600 baud
5	1	4 •	1,200 baud
5	1	5	2,400 baud
5	1	6	4,800 baud
5	1	7	9,600 baud
5	1	8	19,200 baud

5 2 Parity

5	2	2	Space *
5	2	3 •	Odd
5	2	4	Even

5 3 Number of Stop Bits

5	3	1 •	1 stop bit
5	3	2	2 stop bits

5 4 Handshake Mode

5	4	1	Software handshake
5	4	3 •	Hardware handshake after 1 char.

- = Factory setting
- * = Not available on verified balances
- ** = Only on verified balances

5 5		Communication mode
5 5	1 •	SBI
5 5	2	XBPI
5 5	3	YDPO1IS (strip printer)
5 5	4	YDPO2
5 5	5	YDPO3
5 5	6	YDPO1IS (label printer)

5 6		SBN-Address
5 6	1 •	Address 0
5 6	x	Address 1 bis 31 x= 1 bis 32
5 6	32	Address 31

6 Print in Weighing Mode

6 1		Manual/Auto Print Mode
6 1	1	Manual without stability
6 1	2 •	Manual with stability
6 1	4	Automatic without stability
6 1	5	Automatic at stability
6 1	6	Automatic after weight change ***

6 2		Stop Auto Print
6 2	1	Use print key
6 2	2 •	Not possible

6 3		Time-dependent Auto Print
6 3	1 •	1 display update
6 3	2	2 display updates
6 3	4	10 display updates *
6 3	7	100 display updates *

6 4		Print on Request then Tare
6 4	1 •	Off
6 4	2	On

7 Print in Application Mode

7 1		Printing Application Parameters
7 1	1 •	Off
7 1	2	On, all parameters
7 1	3	On, only main parameters

7 2		Line Format
7 2	1	For raw data
7 2	2 •	For other apps./GLP

7 3		Auto Print for Net-Total
7 3	1 •	Last net value
7 3	2	Tare value

8 Extra Functions

8 1		Menu Mode
8 1	1 •	Change parameters
8 1	2	Read only

8 2		Acoustic Signal
8 2	1 •	On
8 2	2	Off

8 3		Keypad
8 3	1 •	Accessible
8 3	2	Blocked
8 3	3	Alphabetic keys blocks

8 4		External Switch Function
8 4	1 •	Print key
8 4	2	Tare key
8 4	3	CAL/isoTEST key
8 4	4	F function key
8 4	5	CF key
8 4	6	Toggle key
8 4	7	Bar code scanner

8 5		Display Backlighting
8 5	1 •	On
8 5	2	Off
8 5	3	Auto off after 4 minutes

8 6		Power-on mode
8 6	1 •	Off > on / standby
8 6	2	Off > on ¹⁾
8 6	4	Auto on

8 7		Automatic shutoff
8 7	1	After 4 minutes
8 7	2 •	Off

8 8		Control Port Function
8 8	1	Input
8 8	2 •	Output

8 10		ISO/GLP/GMP Printout
8 10	1 •	Off
8 10	2	Only for calibration/adjustment
8 10	3	Always on

9 Reset Menu

9 1		Restore Factory Settings
9 1	1	Restore factory settings
9 1	2 •	Do not restore factory settings
9 1	9	Switch to MP8 (interface)
9 1	10 ²⁾	Reset all parameters (incl. INPUT data)
9 1	11 ²⁾	Restore Laboratory version
9 1	12 ²⁾	Restore Industry version

- = Factory setting
- * = Not available on verified balances
- ** = Only on verified balances
- *** = Auto print when the load change is >50d and stability is set to < 25d
- ²⁾ = Only accessible in the service mode

Service Specifications

Service Specifications Sheet 1

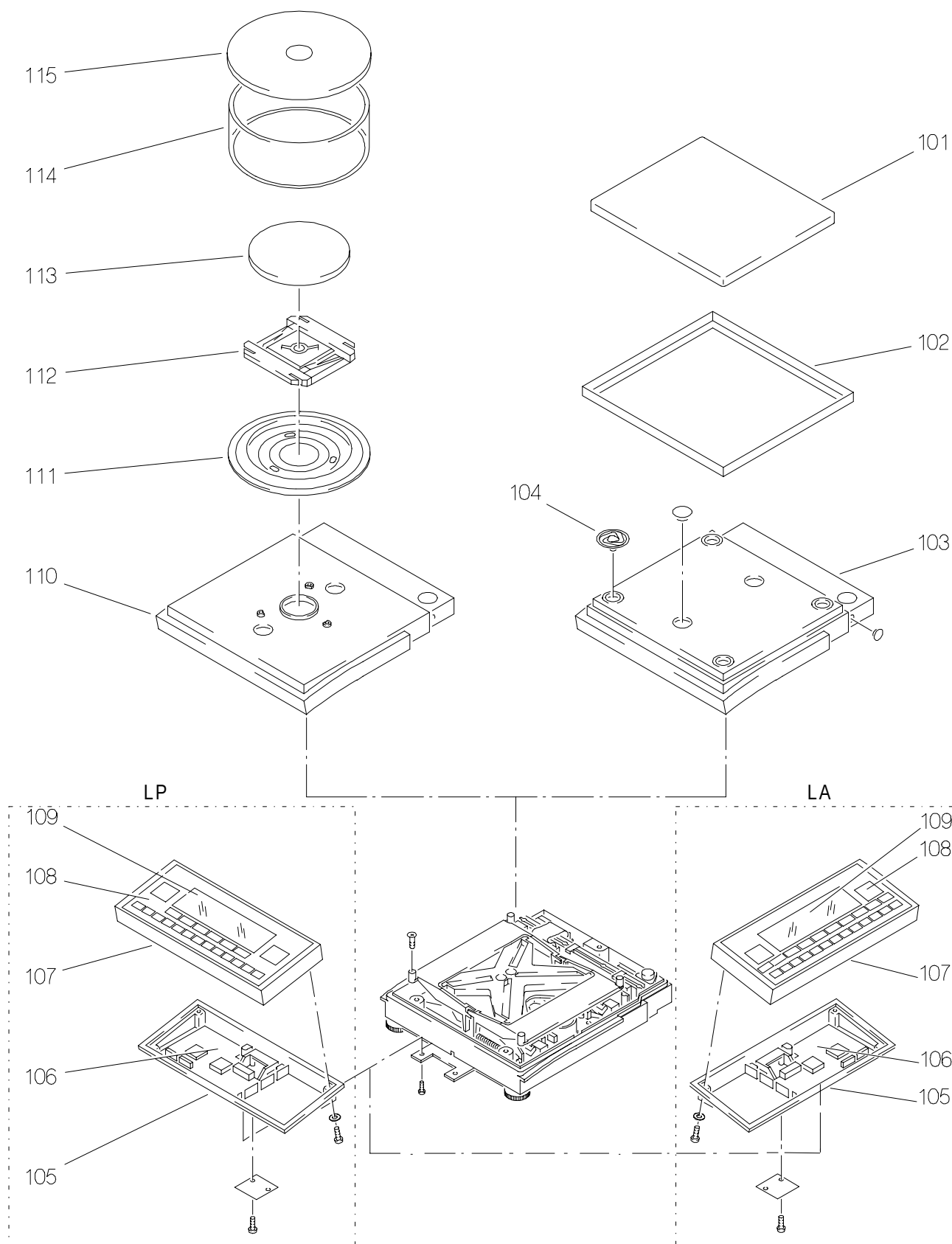
Model	Weighing capacity	Readability [d]	Reproducibility		Off-center load		Class
			Test weight	Permissible tolerances (s)	Test weight	Permissible tolerances (±)	
LP / LA34	34 kg	1 g	10000 g	0,5 g	10000 g	2 g	F2
LP / LA220S	220 g	0,001 g	200 g	0,001 g	200 g	0,005 g	E2
LP / LA420	420 g	0,01 g	200 g	0,01 g	200 g	0,03 g	F2
LP / LA620P	120 240 g 620	0,001 0,002 g 0,005	100 g	0,001 g	500 g	0,007 g	F1
LP / LA620S	620 g	0,001 g	500 g	0,001 g	500 g	0,005 g	E2
LP/ LA820	820 g	0,01 g	500 g	0,01 g	500 g	0,05 g	F1
LP / LA1200S	1200 g	0,001 g	1000 g	0,001 g	1000 g	0,005 g	E2
LP / LA2200P	400 800 g 2200	0,01 0,02 g 0,05	200 g	0,01 g	2000 g	0,07 g	F2
LP / LA2200S	2200 g	0,01 g	2000 g	0,01 g	2000 g	0,06 g	E2
LP / LA2200	2200 g	0,1 g	2000 g	0,05 g	2000 g	0,2 g	F2
LP / LA3200D	1000 3200 g	0,001 0,01 g	1000 g	0,001 g	1000 g	0,005 g	E2
LP / LA4200S	4200 g	0,01 g	2000 g	0,01 g	2000 g	0,04 g	E2
LP / LA4200	4200 g	0,1 g	2000 g	0,05 g	2000 g	0,3 g	F2
LP / LA5200P	1200 2400 g 3800 5200	0,01 0,02 g 0,05 0,1	1000 g	0,01 g	5000 g	0,08 g	F1
LP / LA6200S	6200 g	0,01 g	5000 g	0,01 g	5000 g	0,06 g	E2
LP / LA6200	6200 g	0,1 g	5000 g	0,05 g	5000 g	0,5 g	F1
LP / LA12000P	3000 6000 g 12000	0,1 0,2 g 0,5	2000 g	0,1 g	10000 g	0,7 g	F2
LP / LA12000S	12000 g	0,1 g	10000 g	0,1 g	10000 g	0,6 g	F1
LP / LA16000S	16 kg	0,1 g	10000 g	0,1 g	10000 g	1 g	F1
LP / LA34000P	8 16 kg 34	0,1 0,2 g 0,5	5000 g	0,1 g	10000 g	1 g	F1

Service Specifications Sheet 2

Model	Span accuracy			Linearity			TCS ppm/K
	Adjustment weight	Test weight	Permissible tolerances (±)	Tare weight	Test weight	Permissible tolerances (±)	
LP / LA34	intern	30000 g	1 g	---	7000/15000 22000/30000 g	1 g	1,6
LP / LA220S	intern	200 g	0,002 g	---	50/100/150/200 g	0,002 g	1,8
LP / LA420	intern	400 g	0,02 g	---	100/200/300/400 g	0,01 g	2
LP / LA620P	intern	500 g	0,005 g	100 300 500 g	100 g	0,002 g	1,8
LP / LA620S	intern	500 g	0,004 g	---	150/300/500/600 g	0,002 g	1,8
LP/ LA820	intern	800 g	0,03 g	---	200/400/600/800 g	0,02 g	2
LP / LA1200S	intern	1000 g	0,003 g	200 500 1000 g	200 g	0,002 g	1,8
LP / LA2200P	intern	2000 g	0,05 g	500 1000 1500 g	300 g	0,01 g	2
LP / LA2200S	intern	2000 g	0,04 g	---	500/1000/ 1500/2000 g	0,02 g	2
LP / LA2200	intern	2000 g	0,1 g	---	500/1000/ 1500/2000 g	0,1 g	2
LP / LA3200D	intern	1000 g	0,003 g	---	200/500/700/1000 2000/3000 g	0,002 0,01 g	1,8
LP / LA4200S	intern	4000 g	0,04 g	---	1000/2000/ 3000/4000 g	0,02 g	1,6
LP / LA4200	intern	4000 g	0,2 g	---	1000/2000/ 3000/4000 g	0,1 g	4
LP / LA5200P	intern	5000 g	0,1 g	1000 2500 4000 g	1000 g	0,02 g	4
LP / LA6200S	intern	5000 g	0,04 g	---	1500/3000/ 5000/6000 g	0,02 g	1,8
LP / LA6200	intern	5000 g	0,3 g	---	1500/3000/ 5000/6000 g	0,1 g	4
LP / LA12000P	intern	10000 g	1,0 g	3000 5000 10000 g	2000 g	0,1 g	4
LP / LA12000S	intern	10000 g	0,6 g	---	3000/6000/ 10000/12000 g	0,2 g	4
LP / LA16000S	intern	15000 g	0,3 g	---	4000/8000 12000/16000 g	0,2 g	1,6
LP / LA34000P	intern	30000 g	0,5 g	10000 15000 25000 g	5000 g	0,2 g	1,6

Spare part list LP / LA

LP / LA - Exploded-View Diagram 1



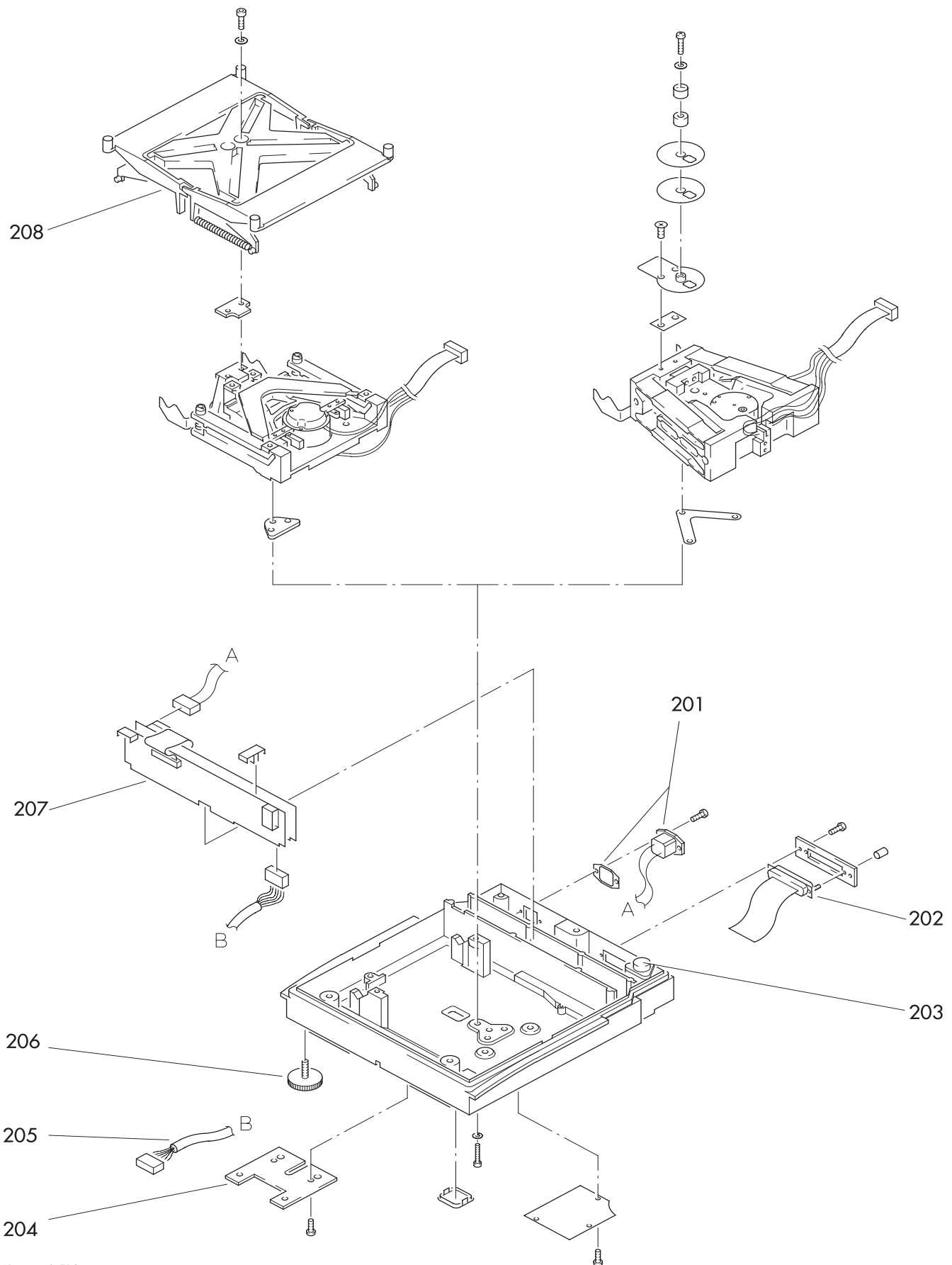
LP_exp_1.WMF

Index	designation	piece	spare part no.	Model LP / LA					
				220S	420 ¹⁾	1200S	4200 ¹⁾	3200D ¹⁾	4200S
				620P	820		6200		5200P
				620S	2200S		12000S		6200S
					2200P		12000P		
					2200				
	AC adapter 6-2, EURO	1	6971886	x	x	x	x	x	x
	AC adapter 6-2, USA	1	6971887	x	x	x	x	x	x
	AC adapter 6-2, GB	1	6971888	x	x	x	x	x	x
101	pan	1	69LP0007		x		x		x
102	frame	1	69LP0008		x				x
103	housing	1	69LP0032		x		x		x
104	shock absorber	4	69LP0010		x		x		x
105	bottom part	1	69LP0025	x	x	x	x	x	x
106 ²⁾	display PCB	1	69LA0005	x	x	x	x	x	x
106	display PCB	1	69LP0022	x	x	x	x	x	x
107	front plate	1	69LA0004	x	x	x	x	x	x
108 ²⁾	switch foil	1	69LA0002	x	x	x	x	x	x
108	switch foil	1	69LP0024	x	x	x	x	x	x
109	glass	1	69LP0023	x	x	x	x	x	x
110	housing	1	69LP0011	x		x		x	
111	shield plate	1	69LP0003	x		x		x	
112	pan support	1	69LP0005	x		x		x	
113	pan	1	69LP0004	x		x		x	
114	glass ring	1	69LP0040	x		x		x	
115	lid	1	69LP0002	x		x		x	
150	sealing, compl.	1	69LP0019	x	x	x	x	x	x
151	set of caps	1	69B20009	x	x	x	x	x	x
152 ²⁾	display	1	69LA0005	x	x	x	x	x	x
152	display	1	69LA0001	x	x	x	x	x	x
154	back light	1	6960LP01	x		x		x	
154	dust cover	1	6960LP02		x		x		x
155	folding box	1	69LP0950	x	x	x	x	x	x
156	pecking, comp.	1	69LP0900	x	x	x	x	x	x

LP_er_de.XLS

¹⁾ = not OEM balances²⁾ = only for LA balances

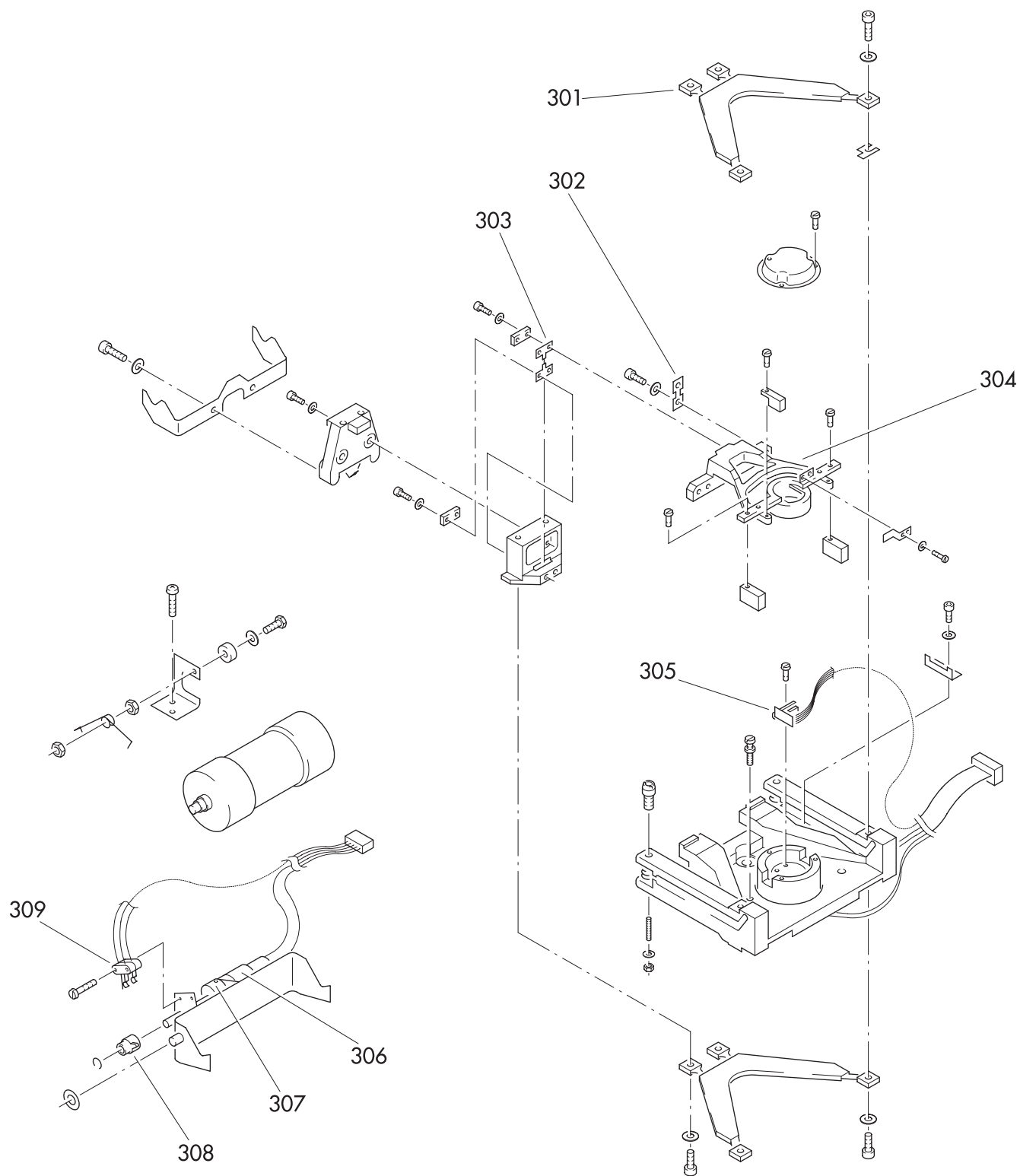
LP / LA - Exploded-View Diagram 2



Index	designation	piece	spare part no.	Model LP / LA					
				220S	420 ¹⁾	1200S	4200 ¹⁾	3200D ¹⁾	4200S
				620P	820		6200		5200P
				620S	2200S		12000S		6200S
					2200P		12000P		
					2200				
201	PC socket	1	69LP0018	x	x	x	x	x	x
202	socket	1	69LP0017	x	x	x	x	x	x
203	spirit level	1	69B12133	x	x	x	x	x	x
204	plate	1	69LP0026	x	x	x	x	x	x
205	cable	1	69LP0027	x	x	x	x	x	x
206	leveling foot	2	69B20005	x	x	x	x	x	x
207	main PCB	1	69LP0035	x	x	x	x	x	x
208	supporting pan	1	69LP0009		x		x		x

LP_er_de.XLS

LP / LA - Exploded-View Diagram 3



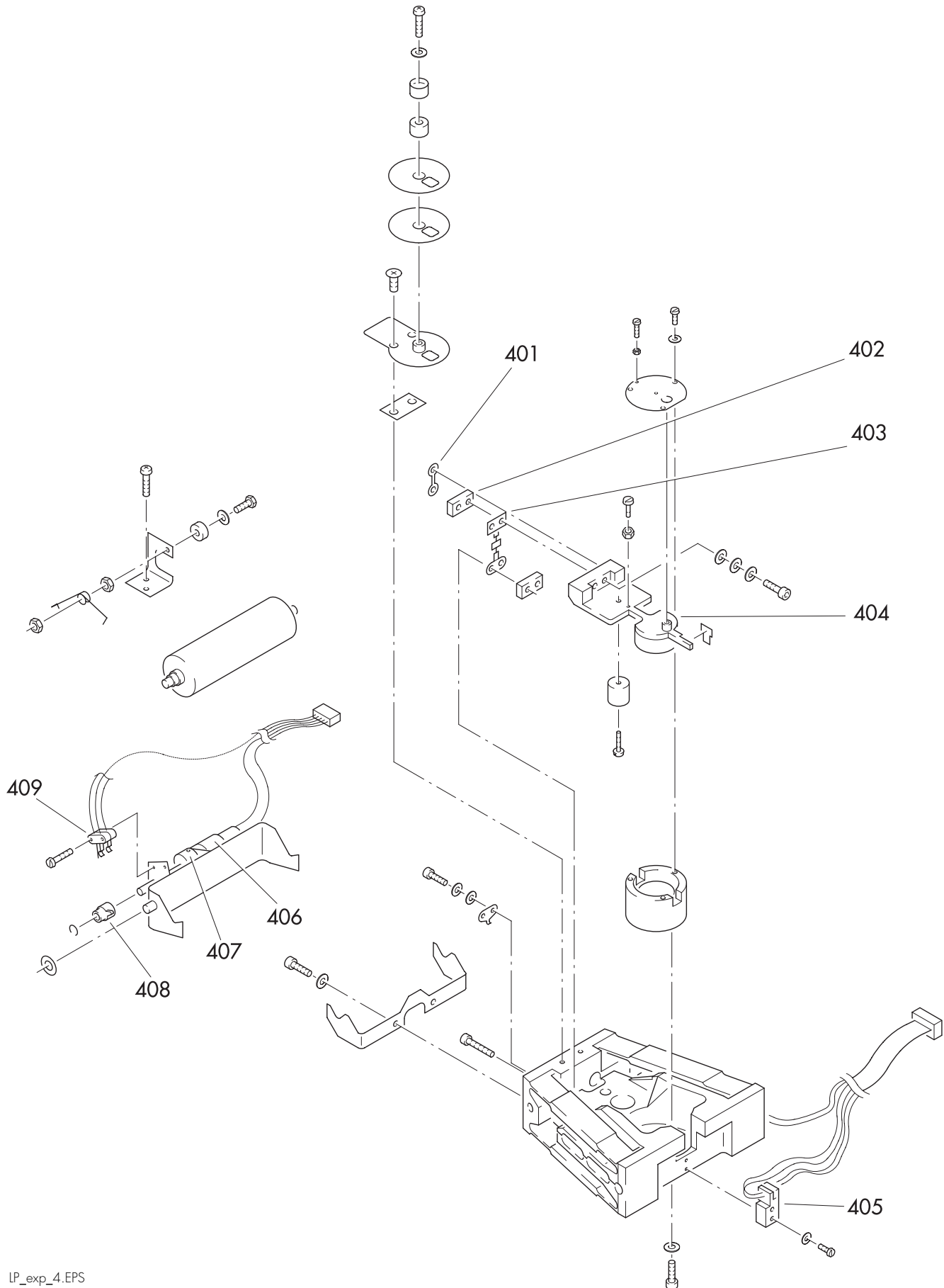
LP_exp_3.EPS

Index	designation	piece	spare part no.			Model LP / LA		
				1200S		3200D ¹⁾		4200S
								5200P
								6200S
301	guide	1	69LC0020	x		x		
301	guide	1	69U61057					x
302	spring	1	69LC0150	x		x		
302	spring	1	69U61060					x
303	coupling band	1	69LC0024	x				
303	coupling band	1	69LC0058					x
303	coupling band	1	69B12043			x		
304	lever, compl.	1	69LP0030	x				
304	lever, compl.	1	69LP0036			x		
304	lever, compl.	1	69B20109					x
305	zero indicator	1	69LP0031	x		x		x
306	motor, compl.	1	69LP0016	x		x		x
307	excenter	1	69LP0015	x		x		x
308	cam	1	69H12223	x		x		x
309	micro switch	2	6918084	x		x		x

LPLA_de.XLS

¹⁾ = not OEM balances

LP / LA - Exploded-View Diagram 4

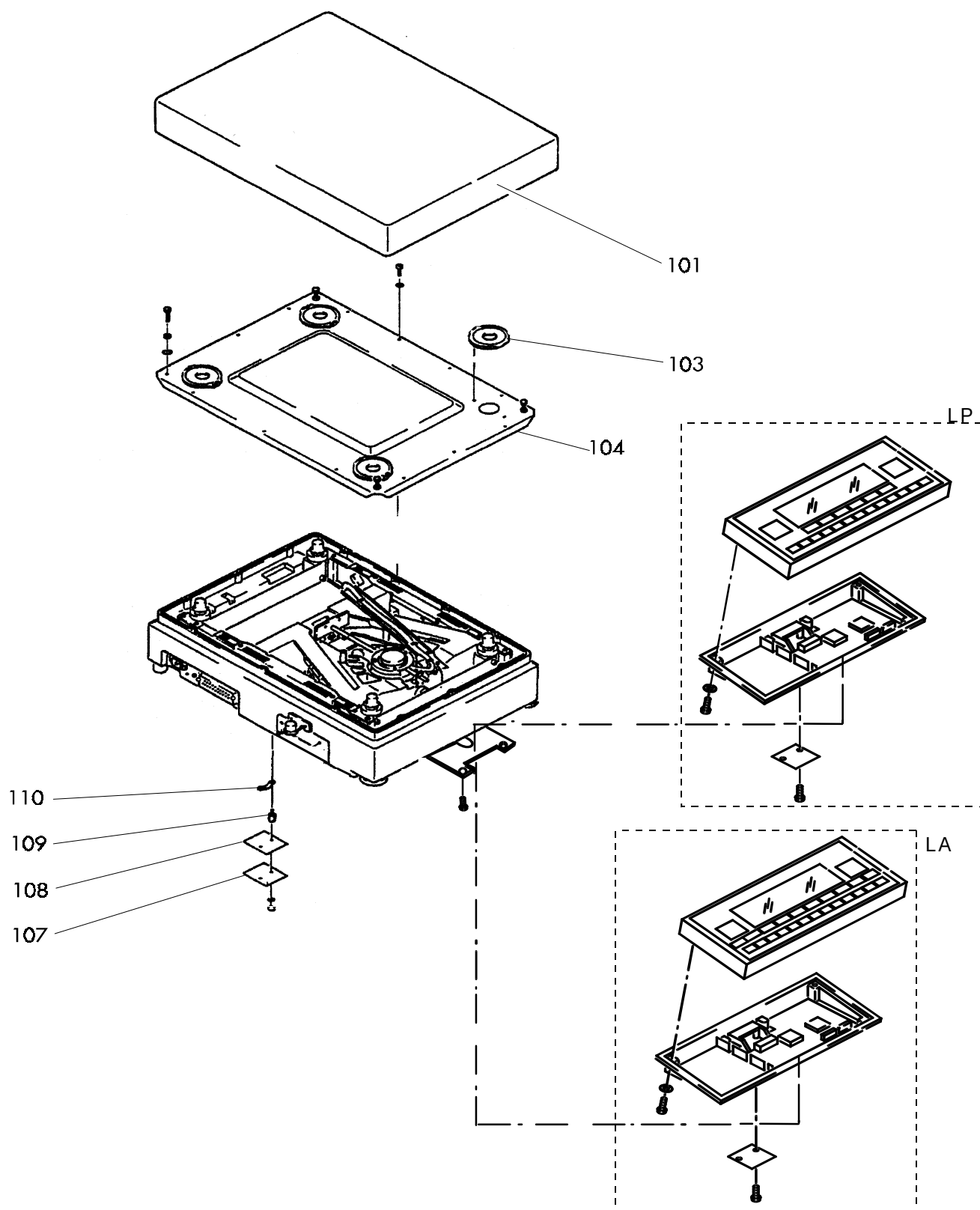


Index	designation	piece	spare part no.		Model LP / LA		
				220S	420 ¹⁾	4200 ¹⁾	
				620P	820	6200	
				620S	2200S	12000S	
					2200P	12000P	
					2200		
401	spring	2	69B20100	x			
401	spring	2	69LC0032		x	x	
402	block	1	69LC0031	x	x	x	
403	coupling band	1	69LC0027	x			
403	coupling band	1	69LC0030		x	x	
404	lever, compl.	1	69B20104	x	x		
404	lever, compl.	1	69B20107			x	
405	null indicator, compl.	1	69LP0014	x	x	x	
406	motor, compl.	1	69LP0016	x	x	x	
407	excenter	1	69LP0015	x	x	x	
408	cam	1	69H12223	x	x	x	
409	micro switch	2	6918084	x	x	x	
450	hook	1	69LC0035	x	x	x	

LPLA_de.XLS

¹⁾ = not OEM balances

LP / LA - Exploded-View Diagram 1a



LP_exp1a.WMF

Index	designation	piece	spare part no.	Model LP / LA					
				16000S		34000P		34	
101	pan	1	69LC0107	x		x		x	
103	protective ring	4	69I31178	x		x		x	
104	cover plate	1	69IS0001	x		x		x	
107	plate	1	69LC0110	x		x		x	
108	seal	1	69LC0111	x		x		x	
109	distance spacer	1	69LC0112	x		x		x	
110	clamp	1	6964989	x		x		x	

LP_er_de.XLS



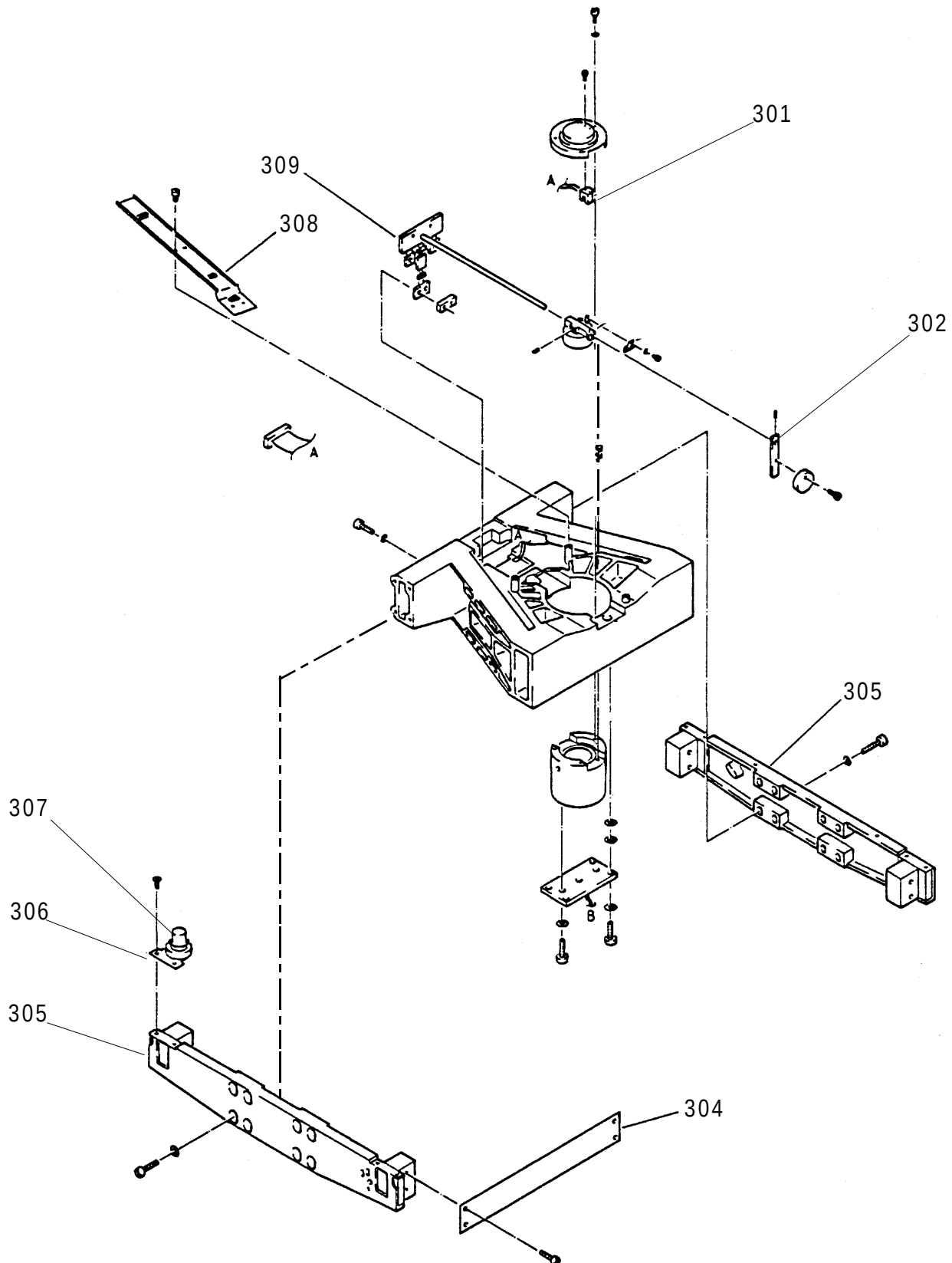
Index	designation	piece	spare part no.	Model LP / LA					
				16000S		34000P		34	
201	buffer	4	69LC0128	x		x		x	
202	levelling foot	3	69LC0093	x		x		x	
203	analog PCB *	1	69LP0038	x		x		x	
203	analog PCB **	1	69LP0041	x		x		x	
206	housing foot	1	69LC0092	x		x		x	
207	guiding device	1	69LC0016	x		x		x	
o.Abb	sealing frame	1	69LI24107	x		x		x	
o.Abb	seal	1	69F50018	x		x		x	
o.Abb	attachment screw	1	69X14228	x		x		x	
208	data output PCB	1	69LP0043	x		x		x	
211	spirit Level	1	6905033	x		x		x	
214	screw	1	69J31113	x		x		x	
215	seling ring	1	6971468	x		x		x	
216	locking screw	1	69LC0117	x		x		x	
217	seal	1	69LC0116	x		x		x	
218	power supply socket	1	69LP0018	x		x		x	

* bis Fab.- Nr.:71007688

** ab Fab.- Nr.:71007689

LP_er_de.XLS

LP / LA - Exploded-View Diagram 3a



LP_exp_3a.WMF

Index	designation	piece	spare part no.	Model LP / LA					
				16000S		34000P		34	
301	Null indicator	1	69LP0042	x		x		x	
301	Null indicator, compl.	1	69IS0007	x		x		x	
302	Block	1	69LC0122	x		x		x	
304	Plate	1	69LC0125	x		x		x	
305	Carrier, left/right	1	69LC0124	x		x		x	
306	Holder	1	69LC0127	x		x		x	
307	Buffer	1	69LC0128	x		x		x	
308	Plate	1	69LC0130	x		x		x	
309	Lever, compl.	1	69IS0008	x		x		x	
				x		x		x	
				x		x		x	
351	Motor, compl.	1	69U61054	x		x		x	
352	Bushing for coupling (shaft)	1	69LC0198	x		x		x	
353	Micro switch	1	6918084	x		x		x	
354	Switch wheel	1	69H12223	x		x		x	
355	Disc for coupling	1	69LC0096	x		x		x	

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